

Sid Mathur's Thematic ETF Screener – Daily Report

2026-05-12

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Action Center

Run date: 2026-05-12 **Universe:** 668 ETFs / 26 clusters

Full PC1: LONG since 2026-04-13 (D+20, 4.0%) **Clusters long:** 14/26 **ETFs long:** 448/668 (67%)

Today's events

BUY (21)

level	id	name	category	cluster_signal	spread_buy	prob_buy	nss
etf	SIVR	abrdrn Physical Silver Shares ...	Silver	FLAT	2.057	0.95	0.796

etf	SLV	iShares Silver Trust	Silver	FLAT	1.939	0.95	0.794
etf	KWEB	KraneShares CSI China Interne...	China Equity	FLAT	0.209	0.94	0.656
etf	LTPZ	Pimco 15 Year U.S. TIPS Index...	Inflation-Protected...	LONG	0.047	0.87	0.292
etf	PPLT	abrdn Physical Platinum Share...	Platinum	FLAT	0.746	0.66	0.155
etf	SILJ	Amplify Junior Silver Miners ...	Materials	FLAT	0.041	0.66	0.141
etf	BAB	Invesco Taxable Municipal Bon...	Taxable Municipal B...	LONG	0.004	0.74	0.098
etf	SIL	Global X Silver Miners ETF	Materials	FLAT	0.735	0.68	0.096
etf	SUB	iShares Short-Term National M...	Municipal Bond	FLAT	0.019	0.90	0.003
etf	VMBS	Vanguard Mortgage-Backed Secu...	Mortgage-Backed Fix...	LONG	0.055	0.78	-0.083
etf	BOND	PIMCO Active Bond Exchange-Tr...	Active Fixed Income	LONG	0.090	0.81	-0.105
etf	MBB	iShares MBS ETF	Mortgage-Backed Fix...	LONG	0.078	0.70	-0.169
cluster	cluster_5	Mixed: BND, AGG, SCHZ	Mixed		0.035	0.70	-0.177
etf	GVI	iShares Intermediate Governme...	Fixed Income	LONG	0.034	0.70	-0.224
etf	AGG	iShares Core U.S. Aggregate B...	Aggregate Fixed Inc...	LONG	0.028	0.65	-0.232
etf	BND	Vanguard Total Bond Market ETF	Aggregate Fixed Inc...	LONG	0.017	0.64	-0.256
etf	SPAB	State Street SPDR Portfolio A...	Aggregate Fixed Inc...	LONG	0.003	0.59	-0.299
etf	SCHZ	Schwab US Aggregate Bond ETF	Aggregate Fixed Inc...	LONG	0.002	0.58	-0.343
etf	NANR	State Street SPDR S&P North A...	Materials	FLAT	0.328	0.35	-0.351
etf	SPMB	State Street SPDR Portfolio M...	Mortgage-Backed Fix...	LONG	0.005	0.53	-0.358
etf	AGZ	iShares Agency Bond ETF	Agency Fixed Income	LONG	0.013	0.45	-0.526

SELL (4)

level	id	name	category	cluster_signal	spread_sell	prob_sell	nss
etf	LVHD	Franklin U.S. Low Volatility ...	U.S. Low Volatility...	LONG	-0.260	0.99	-0.952
etf	FDL	First Trust Morningstar ETF	U.S. Dividend Equity	LONG	-0.277	0.91	-0.883
etf	PKW	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG	-0.691	0.97	0.033
etf	IPAY	Amplify Digital Payments ETF	Financials	FLAT	-0.544	0.82	0.165

Imminent (cluster level, p > 50%)**Imminent BUY (top 5)**

id	name	category	spread_buy	prob_buy	nss
cluster_3	Technology (34/63)	Technology	17.497	1	1.000
cluster_4	Mixed: IWB, ITOT, SCHB	Mixed	8.861	1	0.987
cluster_6	Mixed: IWD, SCHV, VONV	Mixed	4.635	1	0.919
cluster_8	Emerging Markets Equity (19/2...	Emerging Markets Equi...	11.512	1	1.000
cluster_22	Mixed: SRLN, BKLN, FTSL	Mixed	1.305	1	1.000

Imminent SELL (top 5)

id	name	category	spread_sell	prob_sell	nss
cluster_26	Currency (1/2)	Currency	-1.324	1.00	-1.000
cluster_18	Mixed: ILF, EWZ, ECH	Mixed	-0.544	0.91	0.072
cluster_25	India Equity (3/4)	India Equity	-0.556	0.89	0.084
cluster_11	Healthcare (16/17)	Healthcare	-0.362	0.87	0.021
cluster_16	Municipal Bond (13/13)	Municipal Bond	0.032	0.76	0.244

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages (last 12 months).

Current state: LONG since 2026-04-13 (20 days, 4.0%)

5dma: 140.590 20dma: 138.541 50dma: 134.180

Spread vs 50dma: +6.410 Spread vs 20dma: +2.050

P(buy crossover within 5 days): 100.0% P(sell): 4.3%

Top 10 PC1 weights

	ticker	weight	name	category
102	ACWI	0.0021	iShares MSCI ACWI ETF	Global Equity
113	AOA	0.0021	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation
133	SPGM	0.0021	State Street SPDR Portfolio MSCI Global...	Global Equity
137	VT	0.0021	Vanguard Total World Stock Index ETF	Global Equity
46	SPY	0.0020	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity
116	AOR	0.0020	iShares Core 60/40 Balanced Allocation ...	Asset Allocation
120	URTH	0.0020	iShares MSCI World ETF	Global Equity

122	CRBN	0.0020	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity
180	RSP	0.0020	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity
264	IMCG	0.0020	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth

Cluster Overview

change shows the signal-state shift vs the prior run. dNSS is today's NSS minus the prior run's. spark is the last 5 days of NSS (▯ bearish .. ▯ bullish).

	id	name	category	basket	n	signal	change	days	cum_return	nss	dNSS	spark
3	cluster_3	Technology (34/63)	Technology	ONEQ+FXL+ARKW (cor=0.99)	63	FLAT	shifted			1.000	+0.585	▯▯▯▯▯
8	cluster_8	Emerging Markets Equity (19...	Emerging Markets ...	IEMG+DEM+FEMS (cor=1.00)	25	LONG	shifted	21	9.9%	1.000	+0.595	▯▯▯▯▯
22	cluster_22	Mixed: SRLN, BKLN, FTSL	Mixed	SRLN+LQDH+FLOT (cor=0.97)	9	LONG	shifted	24	1.2%	1.000	+0.186	▯▯▯▯▯
23	cluster_23	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	GSY+NEAR+VNLA (cor=0.95)	9	FLAT	shifted			1.000	+0.000	▯▯▯▯▯
4	cluster_4	Mixed: IWB, ITOT, SCHB	Mixed	IWB+VT+IYC (cor=1.00)	61	LONG	shifted	20	5.9%	0.987	+1.183	▯▯▯▯▯
20	cluster_20	Japan Equity (7/9)	Japan Equity	EWJ+DXJ+VPL (cor=1.00)	9	FLAT	shifted			0.947	-0.014	<U+2585><U+2586>
6	cluster_6	Mixed: IWD, SCHV, VONV	Mixed	IWD+CFA+DGRO (cor=1.00)	54	LONG	shifted	20	2.6%	0.919	-0.070	▯▯▯▯▯
21	cluster_21	Broad Commodities (6/9)	Broad Commodities	GSG+DBO+PDBC (cor=1.00)	9	LONG	shifted	84	47.8%	0.918	+0.102	▯▯▯▯▯
24	cluster_24	Currency (4/6)	Currency	IGOV+FXF+FXB (cor=0.98)	6	FLAT	shifted			0.873	-0.127	<U+2587>▯▯▯▯▯
13	cluster_13	China Equity (13/15)	China Equity	CXSE+CNYA+MCHI (cor=0.99)	15	FLAT	shifted			0.858	+1.200	<U+2587>▯▯▯▯▯
2	cluster_2	Mixed: VB, SCHM, IJH	Mixed	VB+PRN+RSPD (cor=1.00)	66	LONG	shifted	20	3.7%	0.849	+0.110	<U+2587>▯▯▯▯▯
14	cluster_14	Real Estate (14/14)	Real Estate	SCHH+RWR+INDS (cor=1.00)	14	LONG	shifted	21	4.2%	0.816	+0.852	▯▯▯▯▯
10	cluster_10	U.S. Dividend Equity (11/20)	U.S. Dividend Equ...	DVY+AML+FDL (cor=0.99)	20	LONG	shifted	14	0.6%	0.787	+0.644	▯▯▯▯▯
19	cluster_19	Inflation-Protected Fixed I...	Inflation-Protect...	TIP+STPZ+SPIP (cor=1.00)	10	LONG	shifted	22	0.8%	0.787	+0.249	▯▯▯▯▯
1	cluster_1	Mixed: VEA, SPDW, SCHF	Mixed	VEA+VGK+VYMI (cor=1.00)	82	FLAT	shifted			0.622	-0.378	<U+2585><U+2586>
12	cluster_12	Materials (13/17)	Materials	GNR+IYM+PICK (cor=0.99)	17	FLAT	shifted			0.550	+0.517	<U+2585><U+2585>
9	cluster_9	High Yield Bond (10) + Pref...	High Yield Bond	JNK+PFF+PCY (cor=0.99)	22	LONG	shifted	20	0.7%	0.489	+1.459	<U+2586><U+2587>
7	cluster_7	Mixed: VBR, MDYV, IVOV	Mixed	VBR+KBE+SPSM (cor=1.00)	48	LONG	shifted	21	2.9%	0.380	-0.619	<U+2587>▯▯<U+2587>
16	cluster_16	Municipal Bond (13/13)	Municipal Bond	VTEB+HYD+CMF (cor=0.99)	13	FLAT	shifted			0.244	-0.051	<U+2586><U+2585>
25	cluster_25	India Equity (3/4)	India Equity	INDA+INDY+EPI+PIN (cor=1.00)	4	FLAT	shifted			0.084	-0.916	<U+2585><U+2585>
18	cluster_18	Mixed: ILF, EWZ, ECH	Mixed	ILF+EIDO+ECH (cor=0.96)	11	FLAT	shifted			0.072	-0.138	___<U+2585><U+2585>
11	cluster_11	Healthcare (16/17)	Healthcare	FXH+XBI+VHT (cor=0.99)	17	LONG	shifted	3	-2.0%	0.021	-0.188	<U+2583><U+2585>
5	cluster_5	Mixed: BND, AGG, SCHZ	Mixed	BND+BLV+BIIV (cor=1.00)	58	LONG	shifted	0	0.0%	-0.177	-1.089	<U+2581><U+2581>
15	cluster_15	Mixed: DBP, GLTR, GDXJ	Mixed	GDXJ+GLTR+SIL (cor=1.00)	13	FLAT	shifted			-0.193	-0.964	<U+2581><U+2581>
17	cluster_17	Consumer Staples (6/11)	Consumer Staples	VDC+FUTY+PSCC (cor=1.00)	11	LONG	shifted	6	-2.3%	-0.232	-1.025	▯▯▯<U+2585>_
26	cluster_26	Currency (1/2)	Currency	SQQQ+UUP (cor=1.00)	2	FLAT	shifted			-1.000	-1.791	<U+2581><U+2581>

Long ETF Positions

ticker	name	category	cluster	cluster_signal	entry_date	days	cum_return	prob_sell	nss
NXTG	First Trust Indxx NextG ETF	Technology	cluster_3	FLAT	2026-04-08	23	21.7%	0.00	1.000
XSD	State Street SPDR S&P Semicondu...	Technology	cluster_3	FLAT	2026-04-09	22	58.0%	0.00	1.000
PTNQ	Pacer Trendpilot 100 ETF	Trend-Following Equ...	cluster_3	FLAT	2026-04-17	16	9.9%	0.00	1.000
VLUE	iShares MSCI USA Value Factor E...	U.S. Value Factor E...	cluster_6	LONG	2026-04-09	22	21.2%	0.00	1.000
FTXL	First Trust Nasdaq Semiconducto...	Technology	cluster_3	FLAT	2026-04-07	24	59.1%	0.00	1.000
EWT	iShares MSCI Taiwan ETF	Taiwan Equity	cluster_8	LONG	2026-04-08	23	27.2%	0.00	1.000
PSI	Invesco Semiconductors ETF	Technology	cluster_3	FLAT	2026-04-07	24	56.9%	0.00	1.000
SOXX	iShares PHLX SOX Semiconductor ...	Technology	cluster_3	FLAT	2026-04-08	23	43.8%	0.00	1.000
SMH	VanEck Semiconductor ETF	Technology	cluster_3	FLAT	2026-04-08	23	36.3%	0.00	1.000
EWY	iShares MSCI South Korea ETF	South Korea Equity	cluster_8	LONG	2026-04-09	22	38.5%	0.00	1.000
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short ...	cluster_23	FLAT	2022-11-11	874	20.4%	0.00	1.000
EWM	iShares MSCI Malaysia Index Fund	Malaysia Equity	cluster_18	FLAT	2026-04-22	13	3.3%	0.00	1.000
FV	First Trust Dorsey Wright Focus...	Momentum Equity	cluster_2	LONG	2026-04-14	19	8.9%	0.00	1.000
AIA	iShares Asia 50 ETF	Asia Equity	cluster_8	LONG	2026-04-10	21	21.0%	0.00	1.000
IXN	iShares Global Tech ETF	Technology	cluster_3	FLAT	2026-04-10	21	24.5%	0.00	1.000
PSCT	Invesco S&P SmallCap Informatio...	U.S. Small Cap Tech...	cluster_3	FLAT	2026-04-07	24	30.0%	0.00	1.000
QQQ	Invesco QQQ Trust, Series 1	U.S. Large Cap Grow...	cluster_3	FLAT	2026-04-10	21	16.7%	0.00	1.000
QTUM	Defiance Quantum ETF	Technology	cluster_3	FLAT	2026-04-10	21	26.7%	0.00	1.000
MTUM	iShares MSCI USA Momentum Facto...	U.S. Momentum Equity	cluster_2	LONG	2026-04-08	23	18.1%	0.00	1.000
PTLC	Pacer Trendpilot US Large Cap E...	U.S. Trend-Followin...	cluster_3	FLAT	2026-04-20	15	4.3%	0.00	1.000
QYLD	Global X NASDAQ 100 Covered Cal...	Options Income	cluster_3	FLAT	2026-04-08	23	4.3%	0.00	1.000
FNDX	Schwab Fundamental U.S. Large C...	U.S. Fundamental Eq...	cluster_6	LONG	2026-04-09	22	5.6%	0.00	1.000
PRF	Invesco RAFI US 1000 ETF	U.S. Fundamental Eq...	cluster_6	LONG	2026-04-09	22	6.4%	0.00	1.000
IWX	iShares Russell Top 200 Value E...	Value Equity	cluster_6	LONG	2026-04-10	21	6.3%	0.00	1.000
PTF	Invesco Dorsey Wright Technolog...	Technology	cluster_3	FLAT	2026-04-07	24	35.1%	0.00	1.000
AAXJ	iShares MSCI All Country Asia e...	Asia ex-Japan Equity	cluster_8	LONG	2026-04-13	20	13.7%	0.00	1.000
EEMA	iShares MSCI Emerging Markets A...	Emerging Markets Eq...	cluster_8	LONG	2026-04-13	20	13.4%	0.00	1.000
BOTZ	Global X Robotics & Artificial ...	Technology	cluster_3	FLAT	2026-04-16	17	12.5%	0.00	1.000
NULV	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-10	21	6.7%	0.00	1.000
EIS	iShares MSCI Israel ETF	Israel Equity	cluster_4	LONG	2026-04-10	21	8.4%	0.00	1.000
XT	iShares Future Exponential Tech...	Technology	cluster_3	FLAT	2026-04-13	20	10.8%	0.00	1.000
EMGF	iShares Emerging Markets Equity...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	13.3%	0.00	1.000
TDIV	First Trust NASDAQ Technology D...	Technology	cluster_3	FLAT	2026-04-13	20	14.3%	0.00	1.000
ROBO	ROBO Global Robotics and Automa...	Technology	cluster_3	FLAT	2026-04-14	19	14.2%	0.00	1.000
XSOE	WisdomTree Emerging Markets Ex...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	12.5%	0.00	1.000
PRN	Invesco Dorsey Wright Industria...	Industrials	cluster_2	LONG	2026-04-08	23	15.4%	0.00	1.000
ONEQ	Fidelity Nasdaq Composite Index...	U.S. Equity	cluster_3	FLAT	2026-04-13	20	13.4%	0.00	1.000
DSI	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equ...	cluster_4	LONG	2026-04-14	19	8.6%	0.00	1.000
ICVT	iShares Convertible Bond ETF	Convertible Bond	cluster_3	FLAT	2026-04-08	23	10.4%	0.00	1.000
EEM	iShares MSCI Emerging Index Fund	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	12.1%	0.00	1.000

JVAL	JPMorgan U.S. Value Factor ETF	U.S. Value Factor E...	cluster_6	LONG	2026-04-10	21	9.1%	0.00	1.000
IOO	iShares Global 100 ETF	Global Equity	cluster_4	LONG	2026-04-13	20	9.1%	0.00	1.000
FPX	First Trust US Equity Opportuni...	IPO Equity	cluster_2	LONG	2026-04-08	23	12.4%	0.00	1.000
IWD	iShares Russell 1000 Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-10	21	5.8%	0.00	1.000
QAI	NYLI Hedge Multi-Strategy Track...	Alternatives	cluster_4	LONG	2026-04-08	23	4.0%	0.00	1.000
VONV	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-10	21	5.8%	0.00	1.000
OEF	iShares S&P 100 Fund	U.S. Mega Cap Equity	cluster_4	LONG	2026-04-13	20	9.5%	0.00	1.000
IEMG	iShares Core MSCI Emerging Mark...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	11.7%	0.00	1.000
ESGE	iShares ESG Aware MSCI EM ETF	Emerging Markets Eq...	cluster_8	LONG	2026-04-13	20	11.2%	0.00	1.000
SUSA	iShares ESG Optimized MSCI USA ...	U.S. ESG Equity	cluster_4	LONG	2026-04-13	20	8.0%	0.00	1.000
PWB	Invesco Large Cap Growth ETF	Growth Equity	cluster_3	FLAT	2026-04-10	21	13.7%	0.00	1.000
SRLN	State Street Blackstone Senior ...	Bank Loan / Floatin...	cluster_22	LONG	2026-04-07	24	1.9%	0.00	1.000
RPG	Invesco S&P 500 Pure Growth ETF	Growth Equity	cluster_2	LONG	2026-04-09	22	14.2%	0.00	1.000
SPHB	Invesco S&P 500 High Beta ETF	U.S. Equity	cluster_3	FLAT	2026-04-10	21	16.4%	0.00	1.000
GEM	Goldman Sachs ActiveBeta Emergi...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	11.3%	0.00	1.000
CWB	State Street SPDR Bloomberg Con...	Convertible Bond	cluster_3	FLAT	2026-04-08	23	10.1%	0.00	1.000
VPL	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	cluster_20	FLAT	2026-04-10	21	10.1%	0.00	1.000
OMFL	Invesco Russell 1000 Dynamic Mu...	U.S. Multifactor Eq...	cluster_4	LONG	2026-04-10	21	7.2%	0.00	1.000
PDP	Invesco Dorsey Wright Momentum ...	Momentum Equity	cluster_2	LONG	2026-04-08	23	9.6%	0.00	1.000
SLX	VanEck Steel ETF	Materials	cluster_12	FLAT	2026-04-09	22	10.8%	0.00	1.000
ESGU	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	cluster_4	LONG	2026-04-13	20	8.1%	0.00	1.000
MGC	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	cluster_4	LONG	2026-04-13	20	8.7%	0.00	1.000
MGV	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	cluster_6	LONG	2026-04-13	20	3.5%	0.00	1.000
GMF	State Street SPDR S&P Emerging ...	Asia Equity	cluster_8	LONG	2026-04-14	19	6.1%	0.00	1.000
DEM	WisdomTree Emerging Markets Hig...	Emerging Markets Eq...	cluster_8	LONG	2026-04-09	22	7.0%	0.00	1.000
EEMV	iShares MSCI Emerging Markets M...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	8.5%	0.00	1.000
FTSL	First Trust Senior Loan Fund	Bank Loan / Floatin...	cluster_22	LONG	2026-04-06	25	1.4%	0.00	0.999
XLP	State Street Consumer Staples S...	Consumer Staples	cluster_17	LONG	2026-05-05	4	-0.8%	0.00	0.999
IVV	iShares Core S&P 500 ETF	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.7%	0.00	0.999
SPY	State Street SPDR S&P 500 ETF T...	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.8%	0.00	0.999
VOO	Vanguard S&P 500 ETF	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.7%	0.00	0.999
ILCV	iShares Morningstar Large-Cap V...	U.S. Large Cap Value	cluster_6	LONG	2026-04-13	20	4.0%	0.00	0.999
VV	Vanguard Large-Cap ETF	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.7%	0.00	0.999
IWL	iShares Russell Top 200 ETF	U.S. Equity	cluster_4	LONG	2026-04-13	20	8.2%	0.00	0.999
ILCB	iShares Morningstar Large-Cap E...	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.7%	0.00	0.999
SPTM	State Street SPDR Portfolio S&P...	U.S. Total Market E...	cluster_4	LONG	2026-04-13	20	7.4%	0.00	0.999
IVW	iShares S&P 500 Growth ETF	U.S. Large Cap Grow...	cluster_3	FLAT	2026-04-13	20	11.5%	0.00	0.999
NURE	Nuveen Short-Term REIT ETF	Real Estate	cluster_14	LONG	2026-04-13	20	5.7%	0.00	0.999
SPYG	State Street SPDR Portfolio S&P...	U.S. Large Cap Grow...	cluster_3	FLAT	2026-04-13	20	11.5%	0.00	0.999
IYY	iShares Dow Jones U.S. ETF	U.S. Total Market E...	cluster_4	LONG	2026-04-13	20	7.5%	0.00	0.999
VOOG	Vanguard S&P 500 Growth ETF	U.S. Large Cap Grow...	cluster_3	FLAT	2026-04-13	20	11.5%	0.00	0.999
IUSG	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Grow...	cluster_3	FLAT	2026-04-13	20	11.2%	0.00	0.999
EPS	WisdomTree U.S. LargeCap Fund	Large Cap Value Equ...	cluster_4	LONG	2026-04-10	21	8.1%	0.00	0.999
SCHB	Schwab U.S. Broad Market ETF	U.S. Total Market E...	cluster_4	LONG	2026-04-13	20	7.4%	0.00	0.999
BKLN	Invesco Senior Loan ETF	Bank Loan / Floatin...	cluster_22	LONG	2026-04-06	25	1.3%	0.00	0.999

ITOT	iShares Core S&P Total U.S. Sto...	U.S. Total Market E...	cluster_4	LONG	2026-04-13	20	7.4%	0.00	0.999
SCHX	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.5%	0.00	0.999
VOOV	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-14	19	3.3%	0.00	0.999
VTI	Vanguard Total Stock Market ETF	U.S. Total Market E...	cluster_4	LONG	2026-04-13	20	7.4%	0.00	0.999
SCHD	Schwab US Dividend Equity ETF	U.S. Dividend Equity	cluster_10	LONG	2026-04-21	14	2.5%	0.00	0.999
SPYX	State Street SPDR S&P 500 Fossi...	U.S. Equity	cluster_4	LONG	2026-04-13	20	7.8%	0.00	0.999
VFMO	Vanguard U.S. Momentum Factor E...	U.S. Momentum Equity	cluster_2	LONG	2026-04-08	23	10.5%	0.00	0.998
XMMO	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Moment...	cluster_2	LONG	2026-04-06	25	13.4%	0.00	0.998
SPYV	State Street SPDR Portfolio S&P...	U.S. Large Cap Value	cluster_6	LONG	2026-04-14	19	3.2%	0.00	0.998
IWB	iShares Russell 1000 ETF	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.2%	0.00	0.998
SPHQ	Invesco S&P 500 Quality ETF	U.S. Quality / Fact...	cluster_4	LONG	2026-04-10	21	5.4%	0.00	0.998
SCHA	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-09	22	8.6%	0.00	0.998
PFXF	VanEck Preferred Securities ex ...	Preferred Stock	cluster_9	LONG	2026-04-14	19	4.1%	0.00	0.998
IVE	iShares S&P 500 Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-14	19	3.2%	0.00	0.998
IWV	iShares Russell 3000 Fund	U.S. Total Market E...	cluster_4	LONG	2026-04-13	20	7.2%	0.00	0.998
VTHR	Vanguard Russell 3000 ETF	U.S. Total Market E...	cluster_4	LONG	2026-04-13	20	7.3%	0.00	0.998
VONE	Vanguard Russell 1000 ETF	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	7.2%	0.00	0.998
SCHV	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-10	21	4.4%	0.00	0.998
CATH	Global X S&P 500 Catholic Value...	Global Equity	cluster_4	LONG	2026-04-13	20	7.3%	0.00	0.997
IUSV	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-14	19	3.1%	0.00	0.997
TILT	FlexShares Mornigstar US Market...	U.S. Multifactor Eq...	cluster_4	LONG	2026-04-13	20	6.0%	0.00	0.996
FTC	First Trust Large Cap Growth Al...	Technology	cluster_2	LONG	2026-04-10	21	9.2%	0.00	0.996
CNYA	iShares MSCI China A ETF	China Equity	cluster_13	FLAT	2026-04-14	19	5.8%	0.00	0.996
IMCG	iShares Morningstar Mid-Cap Gro...	U.S. Mid Cap Growth	cluster_2	LONG	2026-04-09	22	9.1%	0.00	0.996
RWL	Invesco S&P 500 Revenue ETF	U.S. Revenue Weight...	cluster_6	LONG	2026-04-10	21	4.8%	0.01	0.995
ACWI	iShares MSCI ACWI ETF	Global Equity	cluster_4	LONG	2026-04-13	20	6.2%	0.01	0.995
SCHM	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-09	22	7.2%	0.01	0.995
JHML	John Hancock Multifactor Large ...	U.S. Large Cap Equi...	cluster_4	LONG	2026-04-13	20	5.6%	0.01	0.994
SPGM	State Street SPDR Portfolio MSC...	Global Equity	cluster_4	LONG	2026-04-10	21	7.0%	0.01	0.994
VTV	Vanguard Value ETF	U.S. Large Cap Value	cluster_6	LONG	2026-04-13	20	2.8%	0.01	0.993
PSC	Principal U.S. Small-Cap ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-10	21	8.0%	0.01	0.993
VT	Vanguard Total World Stock Inde...	Global Equity	cluster_4	LONG	2026-04-13	20	6.1%	0.01	0.993
ASHR	Xtrackers Harvest CSI 300 China...	China Equity	cluster_13	FLAT	2026-04-13	20	6.9%	0.01	0.993
LRGF	iShares U.S. Equity Factor ETF	U.S. Multifactor Eq...	cluster_4	LONG	2026-04-13	20	6.4%	0.01	0.993
ENZL	iShares MSCI New Zealand ETF	New Zealand Equity	cluster_1	FLAT	2026-05-06	3	0.4%	0.01	0.993
IWM	iShares Russell 2000 Index Fund	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-10	21	9.2%	0.01	0.993
VTWO	Vanguard Russell 2000 ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-10	21	9.2%	0.01	0.993
IWO	iShares Russell 2000 Growth Fund	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-13	20	10.0%	0.01	0.992
QDF	FlexShares Quality Dividend Ind...	Dividend Equity	cluster_4	LONG	2026-04-13	20	5.3%	0.01	0.992
VTWG	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-13	20	10.0%	0.01	0.992
EWI	iShares MSCI Hong Kong Index Fu...	Hong Kong Equity	cluster_13	FLAT	2026-04-08	23	2.3%	0.01	0.991
REZ	iShares Residential and Multise...	Real Estate	cluster_14	LONG	2026-04-14	19	3.8%	0.01	0.991
LQDH	iShares Interest Rate Hedged Co...	Interest Rate Hedge...	cluster_22	LONG	2026-04-06	25	1.3%	0.01	0.990
IGBH	iShares Interest Rate Hedged Lo...	Interest Rate Hedge...	cluster_22	LONG	2026-04-08	23	1.7%	0.01	0.990
CRBN	iShares Low Carbon Optimized MS...	Global Equity	cluster_4	LONG	2026-04-13	20	6.2%	0.01	0.989

AOA	iShares Core 80/20 Aggressive A...	Asset Allocation	cluster_4	LONG	2026-04-13	20	4.8%	0.01	0.989
IWC	iShares Microcap ETF	U.S. Micro Cap Equi...	cluster_2	LONG	2026-04-09	22	10.0%	0.01	0.989
IWY	iShares Russell Top 200 Growth ...	Growth Equity	cluster_3	FLAT	2026-04-14	19	7.7%	0.01	0.988
PDBC	Invesco Optimum Yield Diversifi...	Broad Commodities	cluster_21	LONG	2025-12-22	95	39.8%	0.01	0.988
RDIV	Invesco S&P Ultra Dividend Reve...	U.S. Dividend Equity	cluster_10	LONG	2026-04-21	14	-0.2%	0.01	0.988
GSLC	Goldman Sachs ActiveBeta U.S. L...	Commodity	cluster_4	LONG	2026-04-13	20	6.2%	0.01	0.988
IJK	iShares S&P Mid-Cap 400 Growth ...	U.S. Mid Cap Growth	cluster_2	LONG	2026-04-09	22	6.9%	0.01	0.988
MDYG	State Street SPDR S&P 400 Mid C...	U.S. Mid Cap Growth	cluster_2	LONG	2026-04-09	22	6.9%	0.01	0.987
IWN	iShares Russell 2000 Value ETF	U.S. Small Cap Value	cluster_7	LONG	2026-04-09	22	5.8%	0.01	0.987
SPYD	State Street SPDR Portfolio S&P...	U.S. Dividend Equity	cluster_10	LONG	2026-04-21	14	1.0%	0.01	0.987
KBA	KraneShares Boseri MSCI China A...	China Equity	cluster_13	FLAT	2026-04-10	21	9.8%	0.01	0.987
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	FLAT	2026-04-10	21	17.6%	0.01	0.987
PXH	Invesco RAFI Emerging Markets E...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	5.7%	0.01	0.987
URTH	iShares MSCI World ETF	Global Equity	cluster_4	LONG	2026-04-13	20	5.7%	0.01	0.986
VTWV	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	cluster_7	LONG	2026-04-09	22	5.7%	0.01	0.985
VONG	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Grow...	cluster_3	FLAT	2026-04-14	19	7.3%	0.01	0.985
IWF	iShares Russell 1000 Growth Fund	U.S. Large Cap Grow...	cluster_3	FLAT	2026-04-14	19	7.3%	0.02	0.985
IVOG	Vanguard S&P Mid-Cap 400 Growth...	U.S. Mid Cap Growth	cluster_2	LONG	2026-04-09	22	6.8%	0.02	0.984
DBC	Invesco DB Commodity Index Trac...	Broad Commodities	cluster_21	LONG	2025-12-22	95	38.7%	0.02	0.984
VIDI	Vident International Equity Str...	International Equity	cluster_1	FLAT	2026-04-09	22	6.1%	0.02	0.984
OMFS	Invesco Russell 2000 Dynamic Mu...	U.S. Equity	cluster_7	LONG	2026-04-09	22	6.3%	0.02	0.983
SPEM	State Street SPDR Portfolio Eme...	Emerging Markets Eq...	cluster_8	LONG	2026-04-13	20	5.7%	0.02	0.981
VIOG	Vanguard S&P Small-Cap 600 Grow...	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-08	23	6.6%	0.02	0.980
QMOM	Alpha Architect U.S. Quantitati...	U.S. Momentum Equity	cluster_2	LONG	2026-04-08	23	9.1%	0.02	0.980
ISMD	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap ...	cluster_7	LONG	2026-04-09	22	7.5%	0.02	0.980
KXI	iShares Global Consumer Staples...	Consumer Staples	cluster_17	LONG	2026-05-06	3	-1.6%	0.02	0.980
AOR	iShares Core 60/40 Balanced All...	Asset Allocation	cluster_4	LONG	2026-04-13	20	3.6%	0.02	0.980
SLYG	State Street SPDR S&P 600 Small...	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-08	23	6.6%	0.02	0.979
EES	WisdomTree U.S. SmallCap Fund	Small Cap Value Equ...	cluster_7	LONG	2026-04-08	23	5.1%	0.02	0.978
IJT	iShares S&P SmallCap 600 Growth...	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-08	23	6.6%	0.02	0.978
PAVE	Global X U.S. Infrastructure De...	Industrials	cluster_2	LONG	2026-04-10	21	4.4%	0.02	0.978
XSMO	Invesco S&P SmallCap Momentum E...	U.S. Small Cap Mome...	cluster_2	LONG	2026-04-08	23	7.6%	0.02	0.976
FNDF	Schwab Fundamental Internationa...	International Equity	cluster_1	FLAT	2026-04-08	23	4.2%	0.02	0.976
VWO	Vanguard FTSE Emerging Markets ...	Emerging Markets Eq...	cluster_8	LONG	2026-04-13	20	5.6%	0.02	0.975
VBK	Vanguard Small-Cap Growth ETF	U.S. Small Cap Grow...	cluster_2	LONG	2026-04-10	21	9.9%	0.03	0.975
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short ...	cluster_23	FLAT	2026-04-02	26	0.5%	0.03	0.975
EWJ	iShares MSCI Japan Index Fund	Japan Equity	cluster_20	FLAT	2026-04-13	20	4.4%	0.03	0.974
SCHE	Schwab Emerging Markets Equity ...	Emerging Markets Eq...	cluster_8	LONG	2026-04-13	20	5.2%	0.03	0.974
IYK	iShares U.S. Consumer Staples E...	Consumer Staples	cluster_17	LONG	2026-05-06	3	-0.2%	0.01	0.974
IQDF	FlexShares International Qualit...	International Equity	cluster_1	FLAT	2026-04-10	21	5.0%	0.03	0.973
BBJP	JPMorgan BetaBuilders Japan ETF	Japan Equity	cluster_20	FLAT	2026-04-30	7	3.4%	0.03	0.971
EWX	State Street SPDR S&P Emerging ...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	8.5%	0.03	0.969
PXF	Invesco RAFI Developed Markets ...	International Equity	cluster_1	FLAT	2026-04-08	23	4.2%	0.03	0.969
GWX	State Street SPDR S&P Internati...	International Equity	cluster_1	FLAT	2026-04-13	20	5.1%	0.03	0.968
FNDE	Schwab Fundamental Emerging Mar...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	4.9%	0.03	0.966

XMHQ	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Qualit...	cluster_2	LONG	2026-04-10	21	3.6%	0.03	0.966
XME	State Street SPDR S&P Metals & ...	Materials	cluster_12	FLAT	2026-04-15	18	7.7%	0.04	0.964
VDC	Vanguard Consumer Staples ETF	Consumer Staples	cluster_17	LONG	2026-05-01	6	-1.7%	0.04	0.960
IMCB	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	cluster_6	LONG	2026-04-10	21	5.5%	0.04	0.959
FSTA	Fidelity MSCI Consumer Staples ...	Consumer Staples	cluster_17	LONG	2026-05-01	6	-1.8%	0.04	0.956
VEU	Vanguard FTSE All World Ex US E...	Global ex-US Equity	cluster_1	FLAT	2026-04-10	21	5.0%	0.04	0.956
VI00	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equi...	cluster_7	LONG	2026-04-09	22	5.0%	0.04	0.955
IJR	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equi...	cluster_7	LONG	2026-04-09	22	5.0%	0.05	0.954
PFF	iShares Preferred and Income Se...	Preferred Stock	cluster_9	LONG	2026-04-15	18	1.5%	0.05	0.953
SPSM	State Street SPDR Portfolio S&P...	U.S. Small Cap Equi...	cluster_7	LONG	2026-04-09	22	5.0%	0.05	0.953
HFXI	NYLI FTSE International Equity ...	International Equity	cluster_1	FLAT	2026-04-09	22	4.7%	0.05	0.952
AOM	iShares Core 40/60 Moderate All...	Asset Allocation	cluster_4	LONG	2026-04-13	20	2.4%	0.05	0.952
PRFZ	Invesco RAFI US 1500 Small-Mid ...	U.S. Small/Mid Cap ...	cluster_7	LONG	2026-04-10	21	6.8%	0.05	0.949
VXUS	Vanguard Total International St...	Global ex-US Equity	cluster_1	FLAT	2026-04-10	21	5.1%	0.05	0.946
ISCG	iShares Morningstar Small-Cap G...	Growth Equity	cluster_2	LONG	2026-04-10	21	7.4%	0.05	0.946
DWAS	Invesco Dorsey Wright SmallCap ...	U.S. Small Cap Mome...	cluster_2	LONG	2026-04-07	24	13.7%	0.06	0.943
PSCH	Invesco S&P SmallCap Health Car...	U.S. Small Cap Heal...	cluster_11	LONG	2026-04-13	20	4.8%	0.06	0.943
AMLX	Alerian MLP ETF	MLP / Energy Infrs...	cluster_10	LONG	2026-04-29	8	0.6%	0.01	0.943
IXUS	iShares Core MSCI Total Interna...	Global ex-US Equity	cluster_1	FLAT	2026-04-10	21	5.1%	0.06	0.942
QUS	State Street SPDR MSCI USA Stra...	U.S. Quality / Fact...	cluster_4	LONG	2026-04-14	19	2.5%	0.06	0.941
ARKQ	ARK Autonomous Technology & Rob...	Technology	cluster_3	FLAT	2026-04-14	19	11.1%	0.06	0.941
FEX	First Trust Large Cap Core Alph...	Large Cap Blend Equ...	cluster_6	LONG	2026-04-09	22	4.2%	0.06	0.940
CWI	State Street SPDR MSCI ACWI ex-...	Global ex-US Equity	cluster_1	FLAT	2026-04-10	21	4.6%	0.06	0.939
IWS	iShares Russell Mid-Cap Value E...	U.S. Mid Cap Value	cluster_6	LONG	2026-04-09	22	4.2%	0.06	0.937
SMLF	iShares U.S. Small-Cap Equity F...	U.S. Small Cap Mult...	cluster_2	LONG	2026-04-09	22	6.4%	0.06	0.936
FPXI	First Trust International Equit...	International IPO E...	cluster_4	LONG	2026-04-10	21	7.3%	0.07	0.934
SPDW	State Street SPDR Portfolio Dev...	Developed ex-US Equ...	cluster_1	FLAT	2026-04-10	21	4.6%	0.07	0.933
VXF	Vanguard Extended Market ETF	U.S. Extended Marke...	cluster_2	LONG	2026-04-10	21	7.2%	0.07	0.932
SCHF	Schwab International Equity ETF	Developed ex-US Equ...	cluster_1	FLAT	2026-04-10	21	4.6%	0.07	0.930
AOK	iShares Core 30/70 Conservative...	Asset Allocation	cluster_4	LONG	2026-04-13	20	1.8%	0.07	0.928
ACWX	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	cluster_1	FLAT	2026-04-10	21	4.7%	0.07	0.927
IVOO	Vanguard S&P Mid-Cap 400 ETF		cluster_2	LONG	2026-04-10	21	4.8%	0.07	0.927
SPMD	State Street SPDR Portfolio S&P...	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-10	21	4.8%	0.07	0.926
MDY	State Street SPDR S&P MIDCAP 40...	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-10	21	4.7%	0.07	0.925
IJH	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-10	21	4.7%	0.08	0.924
QUAL	iShares MSCI USA Quality Factor...	U.S. Quality / Fact...	cluster_4	LONG	2026-04-14	19	3.6%	0.08	0.924
IWR	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	cluster_6	LONG	2026-04-10	21	4.9%	0.08	0.924
VEA	Vanguard FTSE Developed Markets...	Developed ex-US Equ...	cluster_1	FLAT	2026-04-10	21	4.5%	0.08	0.921
RPV	Invesco S&P 500 Pure Value ETF	Value Equity	cluster_10	LONG	2026-04-20	15	0.3%	0.08	0.920
VSS	Vanguard FTSE All-Wld ex-US SmC...	Global ex-US Small ...	cluster_1	FLAT	2026-04-13	20	4.8%	0.08	0.916
HEWJ	iShares Currency Hedged MSCI Ja...	Japan Equity	cluster_20	FLAT	2026-04-08	23	2.7%	0.09	0.914
VB	Vanguard Small-Cap ETF	U.S. Small Cap Equi...	cluster_2	LONG	2026-04-10	21	6.1%	0.09	0.914
FEM	First Trust Emerging Markets Al...	Emerging Markets Eq...	cluster_8	LONG	2026-04-07	24	11.5%	0.09	0.914
FNDA	Schwab Fundamental U.S. Small C...	U.S. Small Cap Fund...	cluster_7	LONG	2026-04-10	21	5.4%	0.09	0.911
FXA	Invesco CurrencyShares Australi...	Currency	cluster_1	FLAT	2026-04-13	20	2.1%	0.10	0.900

PSCI	Invesco S&P SmallCap Industrial...	U.S. Small Cap Indu...	cluster_2	LONG	2026-04-13	20	2.7%	0.10	0.896
USRT	iShares Core U.S. REIT ETF	Real Estate	cluster_14	LONG	2026-04-09	22	5.2%	0.11	0.892
EWN	iShares MSCI Netherlands Index ...	Netherlands Equity	cluster_1	FLAT	2026-04-10	21	4.9%	0.11	0.891
DGRW	WisdomTree U.S. Quality Dividen...	U.S. Dividend Equity	cluster_4	LONG	2026-04-14	19	3.2%	0.11	0.890
FYX	First Trust Small Cap Core Alph...	U.S. Small Cap Equi...	cluster_7	LONG	2026-04-08	23	6.5%	0.11	0.890
ETHO	Amplify Etho Climate Leadership...	U.S. ESG Equity	cluster_6	LONG	2026-04-09	22	5.6%	0.11	0.889
IGHG	ProShares Investment Grade-Inte...	Interest Rate Hedge...	cluster_22	LONG	2026-04-07	24	1.1%	0.11	0.889
VO	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	cluster_6	LONG	2026-04-13	20	3.3%	0.11	0.886
HYEM	VanEck Emerging Markets High Yi...	Emerging Markets Fi...	cluster_9	LONG	2026-04-13	20	1.3%	0.12	0.885
SIZE	iShares MSCI USA Size Factor ETF	U.S. Size Factor Eq...	cluster_6	LONG	2026-04-13	20	3.4%	0.12	0.885
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_24	FLAT	2026-05-05	4	0.6%	0.12	0.884
DTD	WisdomTree U.S. Total Dividend ...	U.S. Dividend Equity	cluster_6	LONG	2026-04-13	20	2.5%	0.12	0.881
AIVL	WisdomTree U.S. AI Enhanced Val...	U.S. Value Factor E...	cluster_6	LONG	2026-04-10	21	3.2%	0.12	0.881
DLN	WisdomTree U.S. LargeCap Divide...	U.S. Dividend Equity	cluster_6	LONG	2026-04-13	20	2.6%	0.12	0.879
DIV	Global X Super Dividend ETF	U.S. Dividend Equity	cluster_10	LONG	2026-04-07	24	1.3%	0.12	0.879
PSR	Invesco Active U.S. Real Estate...	Real Estate	cluster_14	LONG	2026-04-09	22	4.9%	0.12	0.878
TDDT	FlexShares iBoxx 3 Year Target ...	Inflation-Protected...	cluster_19	LONG	2026-04-01	27	1.2%	0.12	0.878
FNX	First Trust Mid Cap Core AlphaD...	U.S. Mid/Small Cap ...	cluster_7	LONG	2026-04-10	21	3.9%	0.12	0.877
FXB	Invesco CurrencyShares British ...	Currency	cluster_24	FLAT	2026-04-13	20	1.0%	0.13	0.871
DBJP	Xtrackers MSCI Japan Hedged Equ...	Japan Equity	cluster_20	FLAT	2026-04-08	23	2.5%	0.13	0.870
IPKW	Invesco International BuyBack A...	International Equity	cluster_1	FLAT	2026-04-09	22	2.5%	0.13	0.868
FNDC	Schwab Fundamental Internationa...	International Equity	cluster_1	FLAT	2026-04-10	21	3.7%	0.13	0.867
STPZ	PIMCO 1-5 Year U.S. TIPS Index ...	Inflation-Protected...	cluster_19	LONG	2026-04-02	26	0.9%	0.14	0.862
DGS	WisdomTree Emerging Market Smal...	Emerging Markets Eq...	cluster_8	LONG	2026-04-10	21	4.0%	0.15	0.853
RWJ	Invesco S&P SmallCap 600 Revenu...	U.S. Small Cap Equi...	cluster_7	LONG	2026-04-09	22	4.6%	0.15	0.848
SCZ	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equ...	cluster_1	FLAT	2026-04-10	21	3.7%	0.16	0.844
RWR	State Street SPDR Dow Jones REI...	Real Estate	cluster_14	LONG	2026-04-10	21	4.4%	0.16	0.839
SLYV	State Street SPDR S&P 600 Small...	U.S. Small Cap Value	cluster_7	LONG	2026-04-10	21	4.5%	0.16	0.839
IJS	iShares S&P SmallCap 600 Value ...	U.S. Small Cap Value	cluster_7	LONG	2026-04-10	21	4.5%	0.17	0.831
JETS	U.S. Global Jets ETF	Industrials	cluster_2	LONG	2026-05-06	3	-2.3%	0.17	0.827
RFV	Invesco S&P MidCap 400 Pure Val...	Leveraged / Inverse	cluster_7	LONG	2026-04-10	21	2.9%	0.17	0.826
MLPA	Global X MLP ETF	MLP / Energy Infras...	cluster_10	LONG	2026-04-29	8	-1.5%	0.03	0.825
EWJV	iShares MSCI Japan Value ETF	Japan Equity	cluster_20	FLAT	2026-05-06	3	-0.1%	0.17	0.825
VIOV	Vanguard S&P Small-Cap 600 Valu...	U.S. Small Cap Value	cluster_7	LONG	2026-04-10	21	4.5%	0.18	0.820
SCHH	Schwab U.S. REIT ETF	Real Estate	cluster_14	LONG	2026-04-09	22	4.4%	0.19	0.815
TDTF	FlexShares iBoxx 5 Year Target ...	Inflation-Protected...	cluster_19	LONG	2026-04-09	22	1.0%	0.19	0.812
IGOV	iShares International Treasury ...	International Fixed...	cluster_24	FLAT	2026-04-17	16	-0.1%	0.19	0.812
ICF	iShares Select U.S. REIT ETF	Real Estate	cluster_14	LONG	2026-04-08	23	5.6%	0.19	0.806
RSP	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight E...	cluster_6	LONG	2026-04-14	19	2.2%	0.20	0.805
SCHC	Schwab International Small-Cap ...	Developed ex-US Equ...	cluster_1	FLAT	2026-04-13	20	3.4%	0.20	0.803
DLS	WisdomTree International SmallC...	International Equity	cluster_1	FLAT	2026-04-13	20	2.2%	0.20	0.798
SIVR	abrdn Physical Silver Shares ETF	Silver	cluster_15	FLAT	2026-05-11	0	0.0%	0.15	0.796
SLV	iShares Silver Trust	Silver	cluster_15	FLAT	2026-05-11	0	0.0%	0.15	0.794
TIPX	State Street SPDR Bloomberg 1a?...	Inflation-Protected...	cluster_19	LONG	2026-04-07	24	0.8%	0.21	0.788
TIP	iShares TIPS Bond ETF	Inflation-Protected...	cluster_19	LONG	2026-04-10	21	0.8%	0.22	0.777

DXJ	WisdomTree Japan Hedged Equity ...	Japan Equity	cluster_20	FLAT	2026-05-01	6	4.2%	0.23	0.773
EZM	WisdomTree U.S. MidCap Fund	Mid Cap Value Equit...	cluster_7	LONG	2026-04-10	21	3.7%	0.23	0.771
SCHP	Schwab U.S. TIPS ETF	Inflation-Protected...	cluster_19	LONG	2026-04-13	20	0.6%	0.23	0.769
VIS	Vanguard Industrials ETF	Industrials	cluster_2	LONG	2026-04-13	20	2.1%	0.24	0.764
JHMM	John Hancock Multifactor Mid Ca...	U.S. Mid Cap Equity	cluster_2	LONG	2026-04-10	21	3.9%	0.24	0.762
VIG	Vanguard Div Appreciation ETF	U.S. Dividend Equity	cluster_6	LONG	2026-04-14	19	2.0%	0.25	0.751
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	FLAT	2026-04-10	21	6.8%	0.25	0.748
FIDU	Fidelity MSCI Industrials Index...	Industrials	cluster_2	LONG	2026-04-13	20	2.1%	0.26	0.741
IDV	iShares International Select Di...	International Divid...	cluster_1	FLAT	2026-04-06	25	4.4%	0.26	0.741
PFM	Invesco Dividend Achievers ETF	U.S. Dividend Equity	cluster_6	LONG	2026-04-14	19	1.9%	0.26	0.739
FREL	Fidelity MSCI Real Estate Index...	Real Estate	cluster_14	LONG	2026-04-10	21	4.0%	0.26	0.737
SPIP	State Street SPDR Portfolio TIP...	Inflation-Protected...	cluster_19	LONG	2026-04-10	21	0.7%	0.27	0.732
XLRE	State Street Real Estate Select...	Real Estate	cluster_14	LONG	2026-04-09	22	4.3%	0.27	0.731
FXC	Invesco CurrencyShares Canadian...	Currency	cluster_1	FLAT	2026-04-20	15	-0.2%	0.27	0.731
FDVV	Fidelity High Dividend ETF	U.S. Dividend Equity	cluster_6	LONG	2026-04-14	19	2.4%	0.27	0.726
HDV	iShares Core High Dividend ETF	U.S. Dividend Equity	cluster_10	LONG	2026-05-01	6	-1.0%	0.14	0.726
FEP	First Trust Europe AlphaDEX Fund	Europe Equity	cluster_1	FLAT	2026-04-09	22	2.7%	0.28	0.725
NUSC	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG ...	cluster_2	LONG	2026-04-13	20	3.0%	0.28	0.720
SDOG	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	cluster_10	LONG	2026-04-07	24	1.7%	0.28	0.719
EWI	iShares MSCI Italy ETF	Italy Equity	cluster_1	FLAT	2026-04-08	23	4.7%	0.28	0.716
VNQ	Vanguard Real Estate ETF	Real Estate	cluster_14	LONG	2026-04-10	21	4.0%	0.29	0.710
DIA	State Street SPDR Dow Jones Ind...	U.S. Large Cap Equi...	cluster_6	LONG	2026-04-14	19	2.4%	0.29	0.709
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	LONG	2026-04-07	24	0.4%	0.29	0.708
DHS	WisdomTree U.S. High Dividend F...	U.S. Dividend Equity	cluster_10	LONG	2026-04-20	15	-0.1%	0.30	0.701
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_18	FLAT	2026-04-10	21	3.9%	0.31	0.694
REET	iShares Global REIT ETF	Real Estate	cluster_14	LONG	2026-04-13	20	3.4%	0.32	0.682
XLY	State Street Consumer Discretio...	Consumer Discretion...	cluster_4	LONG	2026-04-15	18	1.0%	0.32	0.679
IYR	iShares U.S. Real Estate ETF	Real Estate	cluster_14	LONG	2026-04-10	21	3.9%	0.32	0.678
VBR	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	cluster_7	LONG	2026-04-10	21	3.3%	0.32	0.676
PEY	Invesco High Yield Equity Divid...	U.S. Dividend Equity	cluster_10	LONG	2026-04-13	20	2.3%	0.33	0.673
EWX	iShares MSCI Mexico ETF	Mexico Equity	cluster_1	FLAT	2026-05-05	4	3.4%	0.33	0.672
XLI	State Street Industrial Select ...	Industrials	cluster_2	LONG	2026-04-13	20	1.3%	0.34	0.664
RWO	State Street SPDR Dow Jones Glo...	Real Estate	cluster_14	LONG	2026-04-13	20	3.0%	0.34	0.662
JPME	JPMorgan Diversified Return U.S...	U.S. Equity	cluster_6	LONG	2026-04-07	24	3.9%	0.34	0.659
FDIS	Fidelity MSCI Consumer Discreti...	Consumer Discretion...	cluster_4	LONG	2026-04-14	19	2.4%	0.34	0.657
URA	Global X Uranium ETF	Materials	cluster_12	FLAT	2026-04-14	19	8.3%	0.35	0.647
VCR	Vanguard Consumer Discretion ETF	Consumer Discretion...	cluster_4	LONG	2026-04-14	19	2.4%	0.35	0.647
BWX	State Street SPDR Bloomberg Int...	International Fixed...	cluster_24	FLAT	2026-05-06	3	-0.2%	0.35	0.646
SPHD	Invesco S&P 500 High Dividend L...	Dividend Equity	cluster_10	LONG	2026-05-05	4	-0.5%	0.30	0.639
VYM	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	cluster_6	LONG	2026-04-10	21	2.5%	0.38	0.618
DES	WisdomTree U.S. SmallCap Divide...	U.S. Small Cap Divi...	cluster_7	LONG	2026-04-08	23	3.1%	0.39	0.613
VNQI	Vanguard Global ex-U.S. Real Es...	Real Estate	cluster_1	FLAT	2026-05-04	5	2.6%	0.41	0.593
FTXR	First Trust Nasdaq Transportati...	Industrials	cluster_2	LONG	2026-04-17	16	-0.5%	0.41	0.592
IMTM	iShares MSCI Intl Momentum Fact...	International Equity	cluster_1	FLAT	2026-04-10	21	2.7%	0.41	0.591
EWG	iShares MSCI Germany Index Fund	Germany Equity	cluster_1	FLAT	2026-04-14	19	0.8%	0.41	0.588

CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fix...	cluster_5	LONG	2026-05-08	1	-0.1%	0.41	0.575
MLPX	Global X MLP & Energy Infrastru...	MLP / Energy Infrast...	cluster_10	LONG	2026-04-30	7	-3.5%	0.07	0.574
EWQ	iShares MSCI Austria ETF	Austria Equity	cluster_1	FLAT	2026-04-10	21	4.2%	0.43	0.571
VIGI	Vanguard International Dividend...	International Divid...	cluster_1	FLAT	2026-04-14	19	0.5%	0.43	0.570
VRP	Invesco Variable Rate Preferred...	Preferred Stock	cluster_9	LONG	2026-04-10	21	0.5%	0.45	0.547
NFRA	FlexShares STOXX Global Broad I...	Global Equity	cluster_1	FLAT	2026-04-09	22	-0.0%	0.46	0.544
SDIV	Global X SuperDividend ETF	Global Dividend Equ...	cluster_8	LONG	2026-04-09	22	1.2%	0.46	0.537
FPE	First Trust Preferred Securitie...	Preferred Stock	cluster_9	LONG	2026-04-14	19	0.6%	0.46	0.536
VYMI	Vanguard International High Div...	International Divid...	cluster_1	FLAT	2026-04-09	22	1.7%	0.46	0.535
IAGG	iShares International Aggregate...	International Fixed...	cluster_5	LONG	2026-05-08	1	-0.3%	0.42	0.533
BNDX	Vanguard Total International Bo...	International Fixed...	cluster_5	LONG	2026-05-08	1	-0.3%	0.41	0.532
OUSA	ALPS O'Shares U.S. Quality Divi...	Dividend Equity	cluster_6	LONG	2026-04-20	15	-0.9%	0.47	0.528
HYD	VanEck High Yield Muni ETF	Municipal Bond	cluster_16	FLAT	2026-04-08	23	0.9%	0.50	0.502
INDS	Pacer Industrial Real Estate ETF	Real Estate	cluster_14	LONG	2026-04-14	19	2.0%	0.51	0.489
RWK	Invesco S&P MidCap 400 Revenue ...	U.S. Mid Cap Equity	cluster_7	LONG	2026-04-09	22	2.7%	0.51	0.486
BBAX	JPMorgan BetaBuilders Developed...	Asia-Pacific Equity	cluster_1	FLAT	2026-04-30	7	0.9%	0.51	0.486
ONEY	State Street SPDR Russell 1000 ...	U.S. Dividend Equity	cluster_10	LONG	2026-04-10	21	2.1%	0.52	0.483
HYMB	State Street SPDR Nuveen ICE Hi...	Municipal Bond	cluster_16	FLAT	2026-04-09	22	0.5%	0.53	0.468
DGRO	iShares Core Dividend Growth ETF	U.S. Dividend Equity	cluster_6	LONG	2026-04-13	20	1.2%	0.54	0.463
DWX	State Street SPDR S&P Internati...	International Divid...	cluster_1	FLAT	2026-05-04	5	1.0%	0.54	0.460
EWL	iShares MSCI Switzerland ETF	Switzerland Equity	cluster_1	FLAT	2026-04-15	18	-0.2%	0.57	0.433
PGX	Invesco Preferred ETF	Preferred Stock	cluster_9	LONG	2026-04-17	16	-0.4%	0.58	0.421
COPX	Global X Copper Miners ETF	Materials	cluster_12	FLAT	2026-05-08	1	1.9%	0.55	0.418
RZV	Invesco S&P SmallCap 600 Pure V...	Leveraged / Inverse	cluster_7	LONG	2026-04-13	20	3.5%	0.58	0.416
PID	Invesco International Dividend ...	International Divid...	cluster_1	FLAT	2026-04-30	7	-0.0%	0.58	0.414
VOX	Vanguard Communication Services...	Communication Servi...	cluster_4	LONG	2026-04-13	20	2.3%	0.60	0.400
MOAT	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	cluster_6	LONG	2026-05-06	3	-0.9%	0.61	0.384
PWV	Invesco Large Cap Value ETF	Value Equity	cluster_6	LONG	2026-04-02	26	3.6%	0.62	0.379
FCOM	Fidelity MSCI Communication Ser...	Communication Servi...	cluster_4	LONG	2026-04-13	20	2.1%	0.62	0.377
PKB	Invesco Building & Construction...	Industrials	cluster_2	LONG	2026-04-13	20	-0.2%	0.63	0.368
HYLS	First Trust Tactical High Yield...	High Yield Bond	cluster_9	LONG	2026-04-09	22	0.5%	0.64	0.363
IYJ	iShares U.S. Industrials ETF	Industrials	cluster_2	LONG	2026-04-13	20	0.7%	0.65	0.353
IJJ	iShares S&P Mid-Cap 400 Value E...	U.S. Mid Cap Value	cluster_7	LONG	2026-04-10	21	2.2%	0.65	0.348
IVOV	Vanguard S&P Mid-Cap 400 Value ...	U.S. Mid Cap Value	cluster_7	LONG	2026-04-10	21	2.1%	0.65	0.347
MDYV	State Street SPDR S&P 400 Mid C...	U.S. Mid Cap Value	cluster_7	LONG	2026-04-10	21	2.2%	0.65	0.346
FMB	First Trust Managed Municipal E...	Municipal Bond	cluster_16	FLAT	2026-04-17	16	0.0%	0.67	0.329
NORW	Global X MSCI Norway ETF	Norway Equity	cluster_12	FLAT	2025-12-09	104	31.8%	0.63	0.320
NYF	iShares New York Muni Bond ETF	Municipal Bond	cluster_16	FLAT	2026-04-15	18	0.4%	0.69	0.310
RDVY	First Trust Rising Dividend Ach...	U.S. Dividend Equity	cluster_6	LONG	2026-04-10	21	3.0%	0.70	0.302
IHDG	WisdomTree International Hedged...	Alternatives	cluster_1	FLAT	2026-05-06	3	-1.5%	0.69	0.297
SPSB	State Street SPDR Portfolio Sho...	Fixed Income	cluster_5	LONG	2026-04-09	22	0.1%	0.71	0.287
CFO	VictoryShares US 500 Enhanced V...	U.S. Multifactor Eq...	cluster_6	LONG	2026-04-13	20	0.7%	0.72	0.278
SDY	State Street SPDR S&P Dividend ...	U.S. Dividend Equity	cluster_6	LONG	2026-05-01	6	-0.9%	0.71	0.277
DON	WisdomTree U.S. MidCap Dividend...	U.S. Mid Cap Divide...	cluster_7	LONG	2026-04-13	20	0.9%	0.72	0.276
PZA	Invesco National AMT-Free Munic...	Municipal Bond	cluster_16	FLAT	2026-04-10	21	0.6%	0.73	0.273

CFA	VictoryShares US 500 Volatility...	U.S. Multifactor Eq...	cluster_6	LONG	2026-04-13	20	0.7%	0.73	0.272
USMV	iShares MSCI USA Min Vol Factor...	U.S. Low Volatility...	cluster_6	LONG	2026-05-05	4	-0.4%	0.64	0.245
ITM	VanEck Intermediate Muni ETF	Municipal Bond	cluster_16	FLAT	2026-05-08	1	0.0%	0.75	0.243
SHYG	iShares 0-5 Year High Yield Cor...	High Yield Bond	cluster_9	LONG	2026-04-07	24	0.9%	0.76	0.243
EQAL	Invesco Russell 1000 Equal Weig...	U.S. Equal Weight E...	cluster_6	LONG	2026-04-08	23	1.5%	0.76	0.238
XLC	State Street Communication Serv...	Communication Servi...	cluster_4	LONG	2026-04-14	19	-0.8%	0.74	0.235
MUB	iShares National Muni Bond ETF	Municipal Bond	cluster_16	FLAT	2026-04-17	16	-0.1%	0.77	0.226
PSCC	Invesco S&P SmallCap Consumer S...	U.S. Small Cap Cons...	cluster_17	LONG	2026-04-28	9	-2.2%	0.77	0.223
RWX	State Street SPDR Dow Jones Int...	Real Estate	cluster_1	FLAT	2026-05-06	3	-1.4%	0.77	0.221
ACWV	iShares MSCI Global Min Vol Fac...	Global Equity	cluster_6	LONG	2026-05-06	3	-0.4%	0.66	0.212
IDLV	Invesco S&P International Devel...	International Equity	cluster_1	FLAT	2026-04-30	7	-0.6%	0.78	0.212
IGLB	iShares 10 Year Investment Grad...	Investment Grade Co...	cluster_5	LONG	2026-05-08	1	-0.4%	0.76	0.196
EWC	iShares MSCI Canada Index Fund	Canada Equity	cluster_1	FLAT	2026-04-09	22	3.0%	0.81	0.187
IYT	iShares U.S. Transportation ETF	Industrials	cluster_2	LONG	2026-04-14	19	1.4%	0.82	0.185
EZA	iShares MSCI South Africa Index...	South Africa Equity	cluster_1	FLAT	2026-05-08	1	0.1%	0.67	0.182
EBND	State Street SPDR Bloomberg Eme...	Emerging Markets Fi...	cluster_1	FLAT	2026-05-06	3	0.0%	0.81	0.180
IYC	iShares U.S. Consumer Discretio...	Consumer Discretion...	cluster_4	LONG	2026-04-14	19	-1.0%	0.82	0.180
FYT	First Trust Small Cap Value Alp...	Value Equity	cluster_7	LONG	2026-04-07	24	4.3%	0.82	0.179
DVY	iShares Select Dividend ETF	U.S. Dividend Equity	cluster_10	LONG	2026-04-08	23	-0.6%	0.82	0.177
USIG	iShares Broad USD Investment Gr...	Investment Grade Co...	cluster_5	LONG	2026-05-06	3	-0.2%	0.81	0.175
VCLT	Vanguard Long-Term Corporate Bo...	Investment Grade Co...	cluster_5	LONG	2026-05-08	1	-0.4%	0.78	0.167
GTO	Invesco Total Return Bond ETF	Active Fixed Income	cluster_5	LONG	2026-05-07	2	0.0%	0.80	0.166
FALN	iShares Fallen Angels USD Bond ...	High Yield Bond	cluster_9	LONG	2026-04-10	21	0.6%	0.84	0.164
FLCO	Franklin Investment Grade Corpo...	Active Fixed Income	cluster_5	LONG	2026-05-07	2	0.3%	0.81	0.162
EWS	iShares MSCI Singapore ETF	Singapore Equity	cluster_4	LONG	2026-04-07	24	1.9%	0.83	0.160
SPLB	State Street SPDR Portfolio Lon...	Investment Grade Co...	cluster_5	LONG	2026-05-08	1	-0.3%	0.78	0.159
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_15	FLAT	2026-05-11	0	0.0%	0.50	0.155
CORP	PIMCO Investment Grade Corporat...	Investment Grade Co...	cluster_5	LONG	2026-05-06	3	-0.2%	0.83	0.153
FXR	First Trust Industrials AlphaDEX	Industrials	cluster_2	LONG	2026-04-14	19	-0.4%	0.85	0.152
ONEV	State Street SPDR Russell 1000 ...	U.S. Equity	cluster_6	LONG	2026-04-14	19	0.3%	0.85	0.152
EMLC	VanEck J. P. Morgan EM Local Cu...	Emerging Markets Fi...	cluster_1	FLAT	2026-05-06	3	-0.0%	0.84	0.151
JPUS	JPMorgan Diversified Return U.S...	U.S. Multifactor Eq...	cluster_6	LONG	2026-04-08	23	0.9%	0.85	0.150
VOE	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	cluster_6	LONG	2026-04-09	22	1.1%	0.85	0.146
SILJ	Amplify Junior Silver Miners ETF	Materials	cluster_15	FLAT	2026-05-11	0	0.0%	0.52	0.141
CMF	iShares California Muni Bond ETF	Municipal Bond	cluster_16	FLAT	2026-05-06	3	0.0%	0.86	0.139
JNK	State Street SPDR Bloomberg Hig...	High Yield Bond	cluster_9	LONG	2026-04-08	23	0.4%	0.87	0.134
CDC	VictoryShares US EQ Income Enha...	U.S. Defensive Equi...	cluster_10	LONG	2026-04-09	22	-0.6%	0.87	0.122
TFI	State Street SPDR Nuveen ICE Mu...	Municipal Bond	cluster_16	FLAT	2026-05-06	3	-0.0%	0.89	0.111
GNMA	iShares GNMA Bond ETF	Mortgage-Backed Fix...	cluster_5	LONG	2026-05-07	2	-0.1%	0.79	0.111
SJNK	State Street SPDR Bloomberg Sho...	High Yield Bond	cluster_9	LONG	2026-04-07	24	0.9%	0.89	0.110
HYS	PIMCO 0-5 Year High Yield Corpo...	High Yield Bond	cluster_9	LONG	2026-04-08	23	0.1%	0.89	0.108
BAB	Invesco Taxable Municipal Bond ...	Taxable Municipal B...	cluster_5	LONG	2026-05-11	0	0.0%	0.65	0.098
BBHY	JPMorgan BetaBuilders USD High ...	High Yield Bond	cluster_9	LONG	2026-04-08	23	0.3%	0.90	0.097
SIL	Global X Silver Miners ETF	Materials	cluster_15	FLAT	2026-05-11	0	0.0%	0.59	0.096
HYG	iShares iBoxx \$ High Yield Corp...	High Yield Bond	cluster_9	LONG	2026-04-08	23	0.3%	0.91	0.090

IGIB	iShares 5-10 Year Investment Gr...	Investment Grade Co...	cluster_5	LONG	2026-05-07	2	0.1%	0.87	0.087
SPIB	State Street SPDR Portfolio Int...	Investment Grade Co...	cluster_5	LONG	2026-05-06	3	-0.1%	0.90	0.081
HYLB	Xtrackers USD High Yield Corpor...	High Yield Bond	cluster_9	LONG	2026-04-08	23	0.2%	0.92	0.077
XSLV	Invesco S&P SmallCap Low Volati...	U.S. Small Cap Low ...	cluster_7	LONG	2026-04-09	22	-0.6%	0.93	0.068
SDVY	First Trust SMID Cap Rising Div...	Dividend Equity	cluster_7	LONG	2026-04-10	21	1.0%	0.93	0.067
LQD	iShares iBoxx \$ Investment Grad...	Investment Grade Co...	cluster_5	LONG	2026-05-08	1	-0.2%	0.85	0.067
RTH	VanEck Retail ETF	Consumer Discretion...	cluster_4	LONG	2026-04-10	21	-0.0%	0.93	0.067
KRE	State Street SPDR S&P Regional ...	Financials	cluster_7	LONG	2026-04-09	22	-1.9%	0.93	0.065
SMDV	ProShares Russell 2000 Dividend...	U.S. Dividend Equity	cluster_7	LONG	2026-04-09	22	-1.0%	0.94	0.061
ANGL	VanEck Fallen Angel High Yield ...	High Yield Bond	cluster_9	LONG	2026-04-09	22	0.2%	0.95	0.053
REM	iShares Mortgage Real Estate ETF	Real Estate	cluster_7	LONG	2026-04-13	20	0.1%	0.95	0.047
AGGY	WisdomTree Yield Enhanced U.S. ...	Aggregate Fixed Inc...	cluster_5	LONG	2026-05-08	1	-0.3%	0.84	0.038
VCIT	Vanguard Intermediate-Term Corp...	Investment Grade Co...	cluster_5	LONG	2026-05-07	2	0.1%	0.90	0.034
KBE	State Street SPDR S&P Bank ETF	Financials	cluster_7	LONG	2026-04-09	22	-1.1%	0.97	0.031
IUSB	iShares Core Universal USD Bond...	Aggregate Fixed Inc...	cluster_5	LONG	2026-05-08	1	-0.2%	0.87	0.020
BSV	Vanguard Short-Term Bond ETF	Fixed Income	cluster_5	LONG	2026-05-06	3	-0.1%	0.91	0.019
HDEF	Xtrackers MSCI EAFE High Divide...	International Equity	cluster_1	FLAT	2026-04-07	24	0.5%	0.93	0.014
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	FLAT	2026-05-11	0	0.0%	0.90	0.003
REGL	ProShares S&P MidCap 400 Divide...	Dividend Equity	cluster_7	LONG	2026-04-10	21	-0.9%	1.00	0.002
QLTA	iShares Aaa A Rated Corporate B...	Investment Grade Co...	cluster_5	LONG	2026-05-08	1	-0.2%	0.90	0.002
IAT	iShares U.S. Regional Banks ETF	Financials	cluster_7	LONG	2026-04-10	21	-1.4%	1.00	0.001
IBB	iShares Biotechnology ETF	Healthcare	cluster_11	LONG	2026-05-06	3	-2.2%	0.81	-0.006
FBND	Fidelity Total Bond ETF	Active Fixed Income	cluster_5	LONG	2026-05-08	1	-0.3%	0.87	-0.047
SMMV	iShares MSCI USA Small-Cap Min ...	U.S. Small Cap Low ...	cluster_7	LONG	2026-04-09	22	-1.2%	1.00	-0.056
VMBS	Vanguard Mortgage-Backed Securi...	Mortgage-Backed Fix...	cluster_5	LONG	2026-05-11	0	0.0%	0.86	-0.083
BOND	PIMCO Active Bond Exchange-Trad...	Active Fixed Income	cluster_5	LONG	2026-05-11	0	0.0%	0.91	-0.105
EWD	iShares MSCI Sweden ETF	Sweden Equity	cluster_1	FLAT	2026-05-06	3	-2.6%	0.96	-0.154
MBB	iShares MBS ETF	Mortgage-Backed Fix...	cluster_5	LONG	2026-05-11	0	0.0%	0.87	-0.169
ECH	iShares MSCI Chile ETF	Chile Equity	cluster_18	FLAT	2026-05-06	3	-2.0%	0.98	-0.170
GVI	iShares Intermediate Government...	Fixed Income	cluster_5	LONG	2026-05-11	0	0.0%	0.93	-0.224
AGG	iShares Core U.S. Aggregate Bon...	Aggregate Fixed Inc...	cluster_5	LONG	2026-05-11	0	0.0%	0.89	-0.232
BND	Vanguard Total Bond Market ETF	Aggregate Fixed Inc...	cluster_5	LONG	2026-05-11	0	0.0%	0.90	-0.256
XHB	State Street SPDR S&P Homebuild...	Consumer Discretion...	cluster_2	LONG	2026-05-06	3	-3.9%	0.98	-0.277
SPAB	State Street SPDR Portfolio Agg...	Aggregate Fixed Inc...	cluster_5	LONG	2026-05-11	0	0.0%	0.89	-0.299
EMLP	First Trust North American Ener...	MLP / Energy Infrac...	cluster_10	LONG	2026-04-28	9	-0.6%	0.56	-0.305
SCHZ	Schwab US Aggregate Bond ETF	Aggregate Fixed Inc...	cluster_5	LONG	2026-05-11	0	0.0%	0.93	-0.343
EWU	iShares MSCI United Kingdom ETF	United Kingdom Equi...	cluster_1	FLAT	2026-04-30	7	-1.1%	0.98	-0.348
IGF	iShares Global Infrastructure E...	Global Equity	cluster_1	FLAT	2026-04-30	7	-1.6%	0.76	-0.350
NANR	State Street SPDR S&P North Ame...	Materials	cluster_12	FLAT	2026-05-11	0	0.0%	0.70	-0.351
SPMB	State Street SPDR Portfolio Mor...	Mortgage-Backed Fix...	cluster_5	LONG	2026-05-11	0	0.0%	0.89	-0.358
FXD	First Trust Cons. Discret. Alph...	Consumer Discretion...	cluster_2	LONG	2026-05-06	3	-3.2%	0.99	-0.388
UUP	Invesco DB USD Index Bullish Fu...	Currency	cluster_26	FLAT	2026-04-27	10	-0.4%	0.47	-0.466
AGZ	iShares Agency Bond ETF	Agency Fixed Income	cluster_5	LONG	2026-05-11	0	0.0%	0.98	-0.526
IGE	iShares North American Natural ...	Energy	cluster_12	FLAT	2026-05-01	6	0.1%	0.64	-0.590
MOO	VanEck Agribusiness ETF	Consumer Staples	cluster_12	FLAT	2026-05-06	3	-2.8%	0.86	-0.816

PHB	cluster_9	LONG	2026-04-10	21
SPLG	cluster_4	LONG	2026-04-13	20
RSX	cluster_18	FLAT	2023-07-27	699

Net Signal Strength (NSS)

NSS = P(buy crossover within 5 days) - P(sell crossover within 5 days). Range [-1, 1]; positive = bullish lean, negative = bearish lean.

Top 20 most bullish (NSS desc)

level	id	name	category	cluster	cluster_signal	prob_buy	prob_sell	nss
etf	NXTG	First Trust Indxx NextG ETF	Technology	cluster_3	FLAT	1	0	1
etf	QTEC	First Trust NASDAQ-100-Technolo...	Technology	cluster_3	FLAT	1	0	1
etf	XSD	State Street SPDR S&P Semicondu...	Technology	cluster_3	FLAT	1	0	1
etf	PTNQ	Pacer Trendpilot 100 ETF	Trend-Following Equ...	cluster_3	FLAT	1	0	1
etf	VLUE	iShares MSCI USA Value Factor E...	U.S. Value Factor E...	cluster_6	LONG	1	0	1
etf	IBDR	iShares iBonds Dec 2026 Term Co...	Target Maturity Bond	cluster_5	LONG	1	0	1
etf	MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short ...	cluster_23	FLAT	1	0	1
etf	FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed...	cluster_22	LONG	1	0	1
etf	ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short ...	cluster_23	FLAT	1	0	1
etf	SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short ...	cluster_23	FLAT	1	0	1
etf	BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short ...	cluster_23	FLAT	1	0	1
etf	GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short ...	cluster_23	FLAT	1	0	1
etf	FTXL	First Trust Nasdaq Semiconducto...	Technology	cluster_3	FLAT	1	0	1
etf	IHF	iShares U.S. Health Care Provid...	Healthcare	cluster_11	LONG	1	0	1
etf	EWT	iShares MSCI Taiwan ETF	Taiwan Equity	cluster_8	LONG	1	0	1
etf	PSI	Invesco Semiconductors ETF	Technology	cluster_3	FLAT	1	0	1
etf	SOXX	iShares PHLX SOX Semiconductor ...	Technology	cluster_3	FLAT	1	0	1
etf	SMH	VanEck Semiconductor ETF	Technology	cluster_3	FLAT	1	0	1
etf	EWY	iShares MSCI South Korea ETF	South Korea Equity	cluster_8	LONG	1	0	1
etf	GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short ...	cluster_23	FLAT	1	0	1

Top 20 most bearish (NSS asc)

level	id	name	category	cluster	cluster_signal	prob_buy	prob_sell	nss
etf	IHI	iShares U.S. Medical Devices ETF	Healthcare	cluster_11	LONG	0.00	1.00	-1.000
cluster	cluster_26	Currency (1/2)	Currency			0.00	1.00	-1.000
etf	SQQQ	ProShares UltraPro Short QQQ	Leveraged / Inverse	cluster_26	FLAT	0.00	1.00	-1.000
etf	SPLV	Invesco S&P 500 Low Volatility ...	U.S. Low Volatility...	cluster_6	LONG	0.00	1.00	-1.000
etf	LGLV	State Street SPDR US Large Cap ...	U.S. Low Volatility...	cluster_6	LONG	0.00	1.00	-0.999
etf	EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_18	FLAT	0.00	1.00	-0.997
etf	FVD	First Trust VL Dividend	U.S. Dividend Equity	cluster_6	LONG	0.00	1.00	-0.996

etf	FTCS	First Trust Capital Strength ETF	U.S. Quality / Fact...	cluster_6	LONG	0.00	1.00	-0.993
etf	IDU	iShares U.S. Utilities ETF	Utilities	cluster_17	LONG	0.00	0.99	-0.992
etf	XLU	State Street Utilities Select S...	Utilities	cluster_17	LONG	0.00	0.99	-0.982
etf	IXJ	iShares Global Healthcare ETF	Healthcare	cluster_11	LONG	0.00	0.98	-0.982
etf	FUTY	Fidelity MSCI Utilities Index E...	Utilities	cluster_17	LONG	0.01	0.99	-0.981
etf	VPU	Vanguard Utilities ETF	Utilities	cluster_17	LONG	0.01	0.99	-0.980
etf	XLV	State Street Health Care Select...	Healthcare	cluster_11	LONG	0.00	0.96	-0.964
etf	LVHD	Franklin U.S. Low Volatility Hi...	U.S. Low Volatility...	cluster_10	LONG	0.04	0.99	-0.952
etf	PPA	Invesco Aerospace & Defense ETF	Industrials	cluster_2	LONG	0.00	0.93	-0.926
etf	IEF	iShares 7-10 Year Treasury Bond...	Fixed Income	cluster_5	LONG	0.04	0.96	-0.926
etf	EDV	Vanguard Extended Duration Trea...	Fixed Income	cluster_5	LONG	0.00	0.92	-0.922
etf	IYH	iShares U.S. Healthcare ETF	Healthcare	cluster_11	LONG	0.01	0.93	-0.920
etf	TLT	iShares 20+ Year Treasury Bond ...	Fixed Income	cluster_5	LONG	0.01	0.93	-0.916

Cluster Details**Cluster 1 – Mixed: VEA, SPDW, SCHF****Members:** 82 **Var explained:** 0.839 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
International Equity	21	0.2686
Developed ex-US Equity	9	0.1179
International Dividend Equity	6	0.0748
Global ex-US Equity	5	0.0656
Europe Equity	4	0.0518
Alternatives	4	0.0464
Eurozone Equity	3	0.0372
Asia-Pacific Equity	2	0.0242
Real Estate	2	0.0225
Global Equity	2	0.0222
Emerging Markets Fixed Income	2	0.0193
Currency	2	0.0181
Commodity	1	0.0132
Unknown	1	0.0130
Global ex-US Small Cap Equity	1	0.0129
Germany Equity	1	0.0124
France Equity	1	0.0124
United Kingdom Equity	1	0.0123
Italy Equity	1	0.0122
Europe Financials	1	0.0120
Sweden Equity	1	0.0120
Netherlands Equity	1	0.0120
Australia Equity	1	0.0119
Spain Equity	1	0.0116
Canada Equity	1	0.0114
Switzerland Equity	1	0.0114
Belgium Equity	1	0.0113
Austria Equity	1	0.0112
Poland Equity	1	0.0104
South Africa Equity	1	0.0099
New Zealand Equity	1	0.0091
Mexico Equity	1	0.0089

Replication basket – VEA+VGK+VYMI (cor=1.00) basket lead at lag 1: -0.001

ticker	weight_ols	name	category	etf_signal
VEA	0.474	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
VGK	0.242	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
VYMI	0.226	Vanguard International High Dividend ...	International Dividen...	LONG

Members

	ticker	weight	name	category	etf_signal
2	EFA	0.0133	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0133	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
63	IEFA	0.0133	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
64	SCHF	0.0133	Schwab International Equity ETF	Developed ex-US Equity	LONG
65	SPDW	0.0133	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
21	DWM	0.0132	WisdomTree International Equity Fund	International Dividend E...	FLAT
34	TLTD	0.0132	FlexShares Morningstar Developed Market...	International Equity	FLAT
49	GSIE	0.0132	Goldman Sachs ActiveBeta International ...	Commodity	FLAT
52	ESGD	0.0132	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
28	PXF	0.0131	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
38	ACWX	0.0131	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
39	CWI	0.0131	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
41	INTF	0.0131	iShares International Equity Factor ETF	International Equity	FLAT
42	IQLT	0.0131	iShares MSCI Intl Quality Factor ETF	International Equity	FLAT
44	QEFA	0.0131	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
58	IEUR	0.0131	iShares Core MSCI Europe ETF	Europe Equity	FLAT
59	IXUS	0.0131	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
60	VEU	0.0131	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
61	VGK	0.0131	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
62	VXUS	0.0131	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
6	IEV	0.0130	iShares Europe ETF	Europe Equity	FLAT
32	EFV	0.0130	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
35	JPIN	0.0130	JPMorgan Diversified Return Internation...	International Equity	FLAT
46	RODM	0.0130	Hartford Multifactor Developed Markets ...	International Equity	FLAT
47	VYMI	0.0130	Vanguard International High Dividend Yi...	International Dividend E...	LONG
48	FNDF	0.0130	Schwab Fundamental International Equity...	International Equity	LONG
56	SCHC	0.0130	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
75	FEU	0.0130			FLAT
24	IQDF	0.0129	FlexShares International Quality Divide...	International Equity	LONG
31	SCZ	0.0129	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG
33	FNDC	0.0129	Schwab Fundamental International Small ...	International Equity	LONG
57	VSS	0.0129	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
8	DLS	0.0128	WisdomTree International SmallCap Fund	International Equity	LONG
19	EZU	0.0128	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT

43	IVLU	0.0128	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
72	HFXI	0.0128	NYLI FTSE International Equity Currency...	International Equity	LONG
29	EFG	0.0127	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
45	FEZ	0.0127	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
54	VIGI	0.0127	Vanguard International Dividend Appreci...	International Dividend E...	LONG
4	FEP	0.0126	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
30	GWX	0.0126	State Street SPDR S&P International Sma...	International Equity	LONG
40	IMTM	0.0126	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
1	EWG	0.0124	iShares MSCI Germany Index Fund	Germany Equity	LONG
5	VIDI	0.0124	Vident International Equity Strategy ETF	International Equity	LONG
16	EWQ	0.0124	iShares MSCI France Index Fund	France Equity	FLAT
20	IDV	0.0124	iShares International Select Dividend E...	International Dividend E...	LONG
53	HDEF	0.0124	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
17	EWU	0.0123	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
50	IDLV	0.0123	Invesco S&P International Developed Low...	International Equity	LONG
79	IPKW	0.0123	Invesco International BuyBack Achievers...	International Equity	LONG
82	EWI	0.0122	iShares MSCI Italy ETF	Italy Equity	LONG
22	EPP	0.0121	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
51	EFAV	0.0121	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
23	EUFN	0.0120	iShares MSCI Europe Financials ETF	Europe Financials	FLAT
71	EWD	0.0120	iShares MSCI Sweden ETF	Sweden Equity	LONG
77	EWN	0.0120	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
81	BBAX	0.0120	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
13	EWA	0.0119	iShares MSCI Australia Index Fund	Australia Equity	FLAT
27	DWX	0.0119	State Street SPDR S&P International Div...	International Dividend E...	LONG
12	HEZU	0.0118	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
36	DBEF	0.0118	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
37	HEFA	0.0118	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
11	PID	0.0117	Invesco International Dividend Achiever...	International Dividend E...	LONG
15	EWI	0.0116	iShares MSCI Spain ETF	Spain Equity	FLAT
9	HEDJ	0.0114	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
10	IHDG	0.0114	WisdomTree International Hedged Quality...	Alternatives	LONG
14	EWL	0.0114	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
18	EWI	0.0114	iShares MSCI Canada Index Fund	Canada Equity	LONG
25	NFRA	0.0114	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG
55	VNQI	0.0114	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
80	EWK	0.0113	iShares MSCI Belgium ETF	Belgium Equity	FLAT
76	EWO	0.0112	iShares MSCI Austria ETF	Austria Equity	LONG
7	RWX	0.0111	State Street SPDR Dow Jones Internation...	Real Estate	LONG
26	IGF	0.0108	iShares Global Infrastructure ETF	Global Equity	LONG
70	EPOL	0.0104	iShares MSCI Poland ETF	Poland Equity	FLAT
74	EZA	0.0099	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
69	FXA	0.0097	Invesco CurrencyShares Australian Dolla...	Currency	LONG
66	EBND	0.0096	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
67	EMLC	0.0096	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG

73	ENZL	0.0091	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
68	EWL	0.0089	iShares MSCI Mexico ETF	Mexico Equity	LONG
78	FXC	0.0084	Invesco CurrencyShares Canadian Dollar ...	Currency	LONG

Cluster 2 – Mixed: VB, SCHM, IJH

Members: 66 **Var explained:** 0.848 **Signal:** LONG since 2026-04-13 (20 days, 3.7%)

Category breakdown

category	n_members	abs_weight
Industrials	15	0.2179
U.S. Small Cap Growth	6	0.0954
U.S. Mid Cap Growth	6	0.0949
Consumer Discretionary	6	0.0845
U.S. Mid Cap Equity	5	0.0813
U.S. Small Cap Equity	5	0.0803
Utilities	3	0.0428
U.S. Momentum Equity	3	0.0428
Growth Equity	2	0.0307
U.S. Small Cap Momentum Equity	2	0.0304
Momentum Equity	2	0.0296
Unknown	1	0.0163
U.S. Small Cap ESG Equity	1	0.0162
U.S. Small Cap Multifactor Equity	1	0.0161
U.S. Extended Market Equity	1	0.0161
U.S. Mid Cap Quality Equity	1	0.0158
U.S. Mid Cap Momentum Equity	1	0.0154
U.S. Small Cap Industrials	1	0.0151
Technology	1	0.0150
U.S. Micro Cap Equity	1	0.0150
IPO Equity	1	0.0143
U.S. Small Cap Consumer Discretionary	1	0.0142

Replication basket – VB+PRN+RSPD (cor=1.00) basket lead at lag 1: -0.005

ticker	weight_ols	name	category	etf_signal
VB	0.687	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG

PRN	0.158	Invesco Dorsey Wright Industrials Mom...	Industrials	LONG
RSPD	0.122	Invesco S&P 500 Equal Weight Consumer...	Consumer Discretionary	FLAT

Members

	ticker	weight	name	category	etf_signal
23	SCHM	0.0163	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
29	MDY	0.0163	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	LONG
37	IVOO	0.0163	Vanguard S&P Mid-Cap 400 ETF		LONG
42	IJH	0.0163	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	LONG
43	SPMD	0.0163	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	LONG
44	VB	0.0163	Vanguard Small-Cap ETF	U.S. Small Cap Equity	LONG
20	JHMM	0.0162	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	LONG
26	NUSC	0.0162	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	LONG
31	IJK	0.0162	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
33	IVOG	0.0162	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
34	MDYG	0.0162	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
45	SCHA	0.0162	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
27	SMLF	0.0161	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
41	VXF	0.0161	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
4	IWM	0.0160	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
30	IJT	0.0160	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	LONG
32	VIOG	0.0160	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	LONG
35	SLYG	0.0160	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	LONG
38	VTWO	0.0160	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
40	ISCG	0.0159	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
15	FXR	0.0158	First Trust Industrials AlphaDEX	Industrials	LONG
28	IWO	0.0158	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
36	VTWG	0.0158	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
39	VBK	0.0158	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
54	PSC	0.0158	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
61	XMHQ	0.0158	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	LONG
14	IMCG	0.0157	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
21	VIS	0.0157	Vanguard Industrials ETF	Industrials	LONG
19	IYJ	0.0156	iShares U.S. Industrials ETF	Industrials	LONG
22	FIDU	0.0156	Fidelity MSCI Industrials Index ETF	Industrials	LONG
60	PAVE	0.0155	Global X U.S. Infrastructure Developmen...	Industrials	LONG
10	XMMO	0.0154	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
11	IWP	0.0153	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	FLAT
13	VOT	0.0153	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	FLAT
46	FXD	0.0153	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
62	VFMO	0.0153	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
66	XSMO	0.0153	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	LONG

1	XLI	0.0152	State Street Industrial Select Sector S...	Industrials	LONG
59	PSCI	0.0151	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	LONG
6	FTC	0.0150	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
24	DWAS	0.0150	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	LONG
25	IWC	0.0150	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
53	RSPD	0.0150	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
7	PDP	0.0149	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
9	RPG	0.0148	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
18	FIW	0.0148	First Trust Water ETF	Utilities	FLAT
48	FTXR	0.0148	First Trust Nasdaq Transportation ETF	Industrials	LONG
5	FV	0.0147	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
63	PKB	0.0147	Invesco Building & Construction ETF	Industrials	LONG
64	PRN	0.0146	Invesco Dorsey Wright Industrials Momen...	Industrials	LONG
16	PHO	0.0145	Invesco Water Resources ETF	Utilities	FLAT
8	FPX	0.0143	First Trust US Equity Opportunities ETF	IPO Equity	LONG
57	XTN	0.0143	State Street SPDR S&P Transportation ETF	Industrials	FLAT
47	PEJ	0.0142	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
65	PSCD	0.0142	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
52	IYT	0.0141	iShares U.S. Transportation ETF	Industrials	LONG
2	XRT	0.0139	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
58	QMOM	0.0138	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
12	MTUM	0.0137	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
56	XHB	0.0137	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
17	CGW	0.0135	Invesco S&P Global Water Index ETF	Utilities	FLAT
55	XAR	0.0135	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
49	PPA	0.0133	Invesco Aerospace & Defense ETF	Industrials	FLAT
3	JETS	0.0127	U.S. Global Jets ETF	Industrials	LONG
50	ITA	0.0125	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT
51	ITB	0.0124	iShares U.S. Home Construction ETF	Consumer Discretionary	FLAT

Cluster 3 – Technology (34/63)

Members: 63 **Var explained:** 0.840 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Technology	34	0.5348
U.S. Large Cap Growth	11	0.1854
U.S. Equity	2	0.0333

Growth Equity	2	0.0332
Unknown	2	0.0315
Convertible Bond	2	0.0306
Financials	2	0.0298
U.S. Mega Cap Growth	1	0.0167
U.S. Equal Weight Equity	1	0.0167
U.S. Small Cap Technology	1	0.0153
IPO Equity	1	0.0152
Options Income	1	0.0152
Trend-Following Equity	1	0.0148
Consumer Discretionary	1	0.0147
U.S. Trend-Following Equity	1	0.0128

Replication basket – ONEQ+FXL+ARKW (cor=0.99) basket lead at lag 1: -0.007

ticker	weight_ols	name	category	etf_signal
ONEQ	0.527	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
FXL	0.376	First Trust Technology AlphaDEX	Technology	FLAT
ARKW	0.073	ARK Next Generation Internet ETF	Technology	FLAT

Members

	ticker	weight	name	category	etf_signal
14	QQQ	0.0170	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
35	IGM	0.0170	iShares Expanded Tech Sector ETF	Technology	FLAT
40	ONEQ	0.0170	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
50	ILCG	0.0170	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
29	XNTK	0.0169	State Street SPDR NYSE Technology ETF	Technology	FLAT
39	NULG	0.0169	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
43	IWF	0.0169	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
45	VGT	0.0169	Vanguard Information Tech ETF	Technology	FLAT
47	FTEC	0.0169	Fidelity MSCI Information Technology In...	Technology	FLAT
48	VONG	0.0169	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
52	SCHG	0.0169	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
54	VUG	0.0169	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
8	AIQ	0.0168	Global X Artificial Intelligence & Tech...	Technology	FLAT
18	QTEC	0.0168	First Trust NASDAQ-100-Technology Secto...	Technology	FLAT
24	FXL	0.0168	First Trust Technology AlphaDEX	Technology	FLAT
37	IYW	0.0168	iShares U.S. Technology ETF	Technology	FLAT
44	IVW	0.0168	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
51	IUSG	0.0168	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG

17	QQEW	0.0167	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	FLAT
36	IXN	0.0167	iShares Global Tech ETF	Technology	LONG
42	IWY	0.0167	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
46	VOOG	0.0167	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
49	MGK	0.0167	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
53	SPYG	0.0167	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
1	XLK	0.0166	State Street Technology Select Sector S...	Technology	FLAT
21	XT	0.0165	iShares Future Exponential Technologies...	Technology	LONG
34	PWB	0.0165	Invesco Large Cap Growth ETF	Growth Equity	LONG
55	SPHB	0.0162	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
20	TDIV	0.0160	First Trust NASDAQ Technology Dividend ...	Technology	LONG
28	FDN	0.0160	First Trust DJ Internet Index Fund	Technology	FLAT
60	FIVG	0.0160			FLAT
6	BOTZ	0.0159	Global X Robotics & Artificial Intellig...	Technology	LONG
7	ROBO	0.0159	ROBO Global Robotics and Automation Ind...	Technology	LONG
63	QTUM	0.0159	Defiance Quantum ETF	Technology	LONG
27	PNQI	0.0158	Invesco Nasdaq Internet ETF	Technology	FLAT
5	SKYY	0.0157	First Trust Cloud Computing ETF	Technology	FLAT
2	SOXX	0.0156	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
3	SMH	0.0156	VanEck Semiconductor ETF	Technology	LONG
4	IGV	0.0155	iShares Expanded Tech-Software Sector E...	Technology	FLAT
22	XSD	0.0155	State Street SPDR S&P Semiconductor ETF	Technology	LONG
38	CWB	0.0155	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
61	PBS	0.0155			FLAT
62	PTF	0.0155	Invesco Dorsey Wright Technology Moment...	Technology	LONG
13	ARKQ	0.0154	ARK Autonomous Technology & Robotics ETF	Technology	LONG
15	NXTG	0.0154	First Trust Indxx NextG ETF	Technology	LONG
19	PSI	0.0153	Invesco Semiconductors ETF	Technology	LONG
57	PSCT	0.0153	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
26	IPO	0.0152	Renaissance IPO ETF	IPO Equity	FLAT
31	FINX	0.0152	Global X FinTech ETF	Financials	FLAT
33	QYLD	0.0152	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
59	FTXL	0.0152	First Trust Nasdaq Semiconductor ETF	Technology	LONG
12	ARKW	0.0151	ARK Next Generation Internet ETF	Technology	FLAT
41	ICVT	0.0151	iShares Convertible Bond ETF	Convertible Bond	LONG
11	ARKK	0.0148	ARK Innovation ETF	Technology	FLAT
25	CIBR	0.0148	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
32	PTNQ	0.0148	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
23	IBUY	0.0147	Amplify Online Retail ETF	Consumer Discretionary	FLAT
10	HACK	0.0146	Amplify Cybersecurity ETF	Technology	FLAT
30	IPAY	0.0146	Amplify Digital Payments ETF	Financials	FLAT
16	PRNT	0.0144	3D Printing (The) ETF	Technology	FLAT
9	BLOK	0.0137	Amplify Blockchain Technology ETF	Technology	FLAT
56	PTLC	0.0128	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG
58	BLCN	0.0123	Siren NexGen Economy ETF	Technology	FLAT

Cluster 4 – Mixed: IWB, ITOT, SCHB**Members:** 61 **Var explained:** 0.892 **Signal:** LONG since 2026-04-13 (20 days, 5.9%)**Category breakdown**

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1558
U.S. Total Market Equity	7	0.1214
Global Equity	7	0.1191
Consumer Discretionary	6	0.0914
Asset Allocation	4	0.0631
Communication Services	4	0.0592
U.S. Mega Cap Equity	3	0.0510
U.S. Quality / Factor Equity	3	0.0505
U.S. Multifactor Equity	3	0.0504
U.S. ESG Equity	2	0.0345
U.S. Equity	2	0.0345
Unknown	2	0.0312
Commodity	1	0.0173
Large Cap Value Equities	1	0.0172
Large Cap Blend Equities	1	0.0171
Dividend Equity	1	0.0168
U.S. Dividend Equity	1	0.0165
Alternatives	1	0.0149
International IPO Equity	1	0.0133
Israel Equity	1	0.0126
Singapore Equity	1	0.0121

Replication basket – IWB+VT+IYC (cor=1.00) basket lead at lag 1: -0.003

ticker	weight_ols	name	category	etf_signal
IWB	0.636	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
VT	0.233	Vanguard Total World Stock Index ETF	Global Equity	LONG
IYC	0.086	iShares U.S. Consumer Discretionary E...	Consumer Discretionary	LONG

Members

	ticker	weight	name	category	etf_signal
37	IWB	0.0174	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG

49	ITOT	0.0174	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
51	SCHB	0.0174	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
4	SPY	0.0173	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG
12	URTH	0.0173	iShares MSCI World ETF	Global Equity	LONG
30	IWV	0.0173	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
31	IYY	0.0173	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
34	SPYX	0.0173	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
36	ESGU	0.0173	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
42	VTHR	0.0173	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
43	GSLC	0.0173	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
45	VONE	0.0173	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
47	VV	0.0173	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
48	ILCB	0.0173	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
50	IVV	0.0173	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
52	SCHX	0.0173	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
53	SPLG	0.0173			LONG
54	SPTM	0.0173	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
55	VOO	0.0173	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
56	VTI	0.0173	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
26	JHML	0.0172	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
29	SUSA	0.0172	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
38	IWL	0.0172	iShares Russell Top 200 ETF	U.S. Equity	LONG
44	EPS	0.0172	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
46	MGC	0.0172	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
7	ACWI	0.0171	iShares MSCI ACWI ETF	Global Equity	LONG
13	CRBN	0.0171	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
15	VT	0.0171	Vanguard Total World Stock Index ETF	Global Equity	LONG
22	TILT	0.0171	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	LONG
28	DSI	0.0171	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
32	LRGF	0.0171	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
8	AOA	0.0170	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
14	SPGM	0.0170	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
33	OEF	0.0170	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
39	QUAL	0.0170	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
24	QDF	0.0168	FlexShares Quality Dividend Index Fund	Dividend Equity	LONG
40	QUS	0.0168	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	LONG
23	IOO	0.0167	iShares Global 100 ETF	Global Equity	LONG
25	CATH	0.0167	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	XLG	0.0167	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
41	SPHQ	0.0166	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
11	AOR	0.0165	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
27	DGRW	0.0165	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
17	IYC	0.0162	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
60	OMFL	0.0161	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	LONG
21	FDIS	0.0159	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG

18	VCR	0.0158	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG
1	XLY	0.0156	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	AOM	0.0155	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
19	VOX	0.0153	Vanguard Communication Services ETF	Communication Services	LONG
20	FCOM	0.0153	Fidelity MSCI Communication Services In...	Communication Services	LONG
2	XLC	0.0149	State Street Communication Services Sel...	Communication Services	LONG
5	QAI	0.0149	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
59	RTH	0.0147	VanEck Retail ETF	Consumer Discretionary	LONG
9	AOK	0.0142	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
61	IGN	0.0139			FLAT
57	IYZ	0.0137	iShares U.S. Telecommunications ETF	Communication Services	FLAT
16	FPXI	0.0133	First Trust International Equity Opport...	International IPO Equity	LONG
3	ESPO	0.0132	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
58	EIS	0.0126	iShares MSCI Israel ETF	Israel Equity	LONG
6	EWS	0.0121	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 5 – Mixed: BND, AGG, SCHZ

Members: 58 **Var explained:** 0.832 **Signal:** LONG since 2026-05-11 (0 days, 0.0%)

Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3282
Investment Grade Corporate Bond	13	0.2301
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0890
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0309
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND+BLV+BIV (cor=1.00) basket lead at lag 1: 0.001

ticker	weight_ols	name	category	etf_signal
BND	0.505	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

BLV	0.134	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
BIV	0.219	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT

Members

	ticker	weight	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	LONG
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	LONG
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	LONG
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	LONG
46	SCHR	0.0182	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	LONG
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	LONG
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
50	MBB	0.0180	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
52	SPMB	0.0179	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	LONG
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT

16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT
39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	LONG
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0157	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0152	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0136	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	LONG
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	FLAT
32	LDUR	0.0120	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 6 – Mixed: IWD, SCHV, VONV

Members: 54 **Var explained:** 0.922 **Signal:** LONG since 2026-04-13 (20 days, 2.6%)

Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	11	0.2040
U.S. Large Cap Value	10	0.1903
U.S. Multifactor Equity	3	0.0559
U.S. Mid Cap Equity	3	0.0556
U.S. Value Factor Equity	3	0.0553
U.S. Low Volatility Equity	3	0.0514
U.S. Fundamental Equity	2	0.0380
U.S. Equal Weight Equity	2	0.0377
U.S. Mid Cap Value	2	0.0376
U.S. Equity	2	0.0375
Value Equity	2	0.0369

U.S. Revenue Weighted Equity	1	0.0188
Large Cap Blend Equities	1	0.0188
U.S. Mega Cap Value	1	0.0188
U.S. Size Factor Equity	1	0.0187
U.S. Large Cap Equity	1	0.0184
U.S. Buyback Equity	1	0.0183
Dividend Equity	1	0.0180
U.S. Quality / Factor Equity	1	0.0179
U.S. Moat Equity	1	0.0177
U.S. ESG Equity	1	0.0176
Global Equity	1	0.0168

Replication basket – IWD+CFA+DGRO (cor=1.00) basket lead at lag 1: 0.010

ticker	weight_ols	name	category	etf_signal
IWD	0.459	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
CFA	0.310	VictoryShares US 500 Volatility Wtd E...	U.S. Multifactor Equi...	LONG
DGRO	0.222	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG

Members

	ticker	weight	name	category	etf_signal
22	IWD	0.0192	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
2	RSP	0.0191	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
9	PRF	0.0191	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
29	VONV	0.0191	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
32	ILCV	0.0191	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	LONG
35	SCHV	0.0191	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
12	CFA	0.0190	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	LONG
18	FNDX	0.0190	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
24	IVE	0.0190	iShares S&P 500 Value ETF	U.S. Large Cap Value	LONG
25	JPUS	0.0190	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	LONG
34	IUSV	0.0190	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	LONG
38	VTV	0.0190	Vanguard Value ETF	U.S. Large Cap Value	LONG
45	DTD	0.0190	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
50	DGRO	0.0190	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
15	NULV	0.0189	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	LONG
20	IWX	0.0189	iShares Russell Top 200 Value ETF	Value Equity	LONG
28	VOOV	0.0189	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
36	SPYV	0.0189	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	LONG
44	DLN	0.0189	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG

5	FEX	0.0188	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
10	RWL	0.0188	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
30	MGV	0.0188	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
31	VYM	0.0188	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	LONG
39	IWS	0.0188	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
40	VOE	0.0188	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	LONG
42	PFM	0.0188	Invesco Dividend Achievers ETF	U.S. Dividend Equity	LONG
11	AIVL	0.0187	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
21	ONEV	0.0187	State Street SPDR Russell 1000 Low Volatility	U.S. Equity	LONG
26	SIZE	0.0187	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
51	VIG	0.0187	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
53	JPME	0.0187	JPMorgan Diversified Return U.S. Mid Cap	U.S. Equity	LONG
19	EQAL	0.0186	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	LONG
33	IMCB	0.0186	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
54	JVAL	0.0186	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
17	FDVV	0.0185	Fidelity High Dividend ETF	U.S. Dividend Equity	LONG
23	IWR	0.0185	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
37	VO	0.0185	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
47	DIA	0.0184	State Street SPDR Dow Jones Industrial	U.S. Large Cap Equity	LONG
4	PKW	0.0183	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	FLAT
14	NOBL	0.0182	ProShares S&P 500 Dividend Aristocrats	U.S. Dividend Equity	FLAT
6	PWV	0.0181	Invesco Large Cap Value ETF	Value Equity	LONG
7	RDVY	0.0181	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	LONG
3	FVD	0.0180	First Trust VL Dividend	U.S. Dividend Equity	FLAT
16	SDY	0.0180	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
27	VLUE	0.0180	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
43	OUSA	0.0180	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
41	FTCS	0.0179	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	FLAT
49	LGLV	0.0179	State Street SPDR US Large Cap Low Volatility	U.S. Low Volatility Equi...	FLAT
13	CFO	0.0178	VictoryShares US 500 Enhanced Volatility	U.S. Multifactor Equity	LONG
8	MOAT	0.0177	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	LONG
48	USMV	0.0176	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
52	ETHO	0.0176	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	LONG
1	ACWV	0.0168	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG
46	SPLV	0.0159	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT

Cluster 7 – Mixed: VBR, MDYV, IVOV**Members:** 48 **Var explained:** 0.880 **Signal:** LONG since 2026-04-10 (21 days, 2.9%)**Category breakdown**

category	n_members	abs_weight
Financials	13	0.2576
U.S. Small Cap Value	6	0.1304
U.S. Small Cap Equity	5	0.1082
U.S. Mid Cap Value	3	0.0657
Dividend Equity	2	0.0424
Leveraged / Inverse	2	0.0422
U.S. Small Cap Low Volatility	2	0.0407
U.S. Small Cap Fundamental Equity	1	0.0218
Mid Cap Value Equities	1	0.0218
U.S. Mid Cap Dividend Equity	1	0.0217
Small Cap Value Equities	1	0.0216
U.S. Small Cap Dividend Equity	1	0.0216
U.S. Mid/Small Cap Equity	1	0.0216
U.S. Small/Mid Cap Fundamental Equity	1	0.0215
Value Equity	1	0.0215
U.S. Mid Cap Equity	1	0.0214
U.S. Small/Mid Cap Equity	1	0.0213
U.S. Equity	1	0.0210
U.S. Dividend Equity	1	0.0206
U.S. Mid Cap Low Volatility	1	0.0198
Global Equity	1	0.0181
Real Estate	1	0.0173

Replication basket – VBR+KBE+SPSM (cor=1.00) basket lead at lag 1: -0.000

ticker	weight_ols	name	category	etf_signal
VBR	0.557	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	LONG
KBE	0.164	State Street SPDR S&P Bank ETF	Financials	LONG
SPSM	0.222	State Street SPDR Portfolio S&P 600 S...	U.S. Small Cap Equity	LONG

Members

	ticker	weight	name	category	etf_signal
37	VBR	0.0220	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	LONG
28	IJJ	0.0219	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
30	IVOV	0.0219	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	LONG
31	MDYV	0.0219	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	LONG
21	EZM	0.0218	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	LONG
23	FNDA	0.0218	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	LONG

8	FYX	0.0217	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	LONG
10	VI00	0.0217	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	LONG
11	IJR	0.0217	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	LONG
12	SPSM	0.0217	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	LONG
19	DON	0.0217	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	LONG
26	IWN	0.0217	iShares Russell 2000 Value ETF	U.S. Small Cap Value	LONG
29	IJS	0.0217	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
32	SLYV	0.0217	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	LONG
34	VTWV	0.0217	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	LONG
47	SDVY	0.0217	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
7	FNX	0.0216	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	LONG
18	DES	0.0216	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	LONG
20	EES	0.0216	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	LONG
33	VIOV	0.0216	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	LONG
9	PRFZ	0.0215	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	LONG
46	FYT	0.0215	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	LONG
41	RWK	0.0214	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	LONG
38	RWJ	0.0213	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	LONG
48	ISMD	0.0213	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
39	RFV	0.0212	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	LONG
42	OMFS	0.0210	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	LONG
13	FXO	0.0209	First Trust Financials AlphaDEX	Financials	FLAT
43	RZV	0.0209	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	LONG
6	REGL	0.0207	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
17	SMDV	0.0206	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	LONG
25	XSLV	0.0204	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	LONG
35	VFH	0.0204	Vanguard Financials ETF	Financials	FLAT
36	FNCL	0.0204	Fidelity MSCI Financials Index ETF	Financials	FLAT
27	SMMV	0.0203	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
3	KBE	0.0202	State Street SPDR S&P Bank ETF	Financials	LONG
15	IYF	0.0202	iShares U.S. Financial ETF	Financials	FLAT
45	KCE	0.0202	State Street SPDR S&P Capital Markets E...	Financials	FLAT
16	IYG	0.0201	iShares U.S. Financial Services ETF	Financials	FLAT
24	XMLV	0.0198	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
1	XLF	0.0197	State Street Financial Select Sector SP...	Financials	FLAT
22	KBWB	0.0197	Invesco KBW Bank ETF	Financials	FLAT
2	KRE	0.0196	State Street SPDR S&P Regional Banking ...	Financials	LONG
14	IAT	0.0194	iShares U.S. Regional Banks ETF	Financials	LONG
44	IAI	0.0189	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
40	PSP	0.0181	Invesco Global Listed Private Equity ETF	Global Equity	FLAT
4	KIE	0.0177	State Street SPDR S&P Insurance ETF	Financials	FLAT
5	REM	0.0173	iShares Mortgage Real Estate ETF	Real Estate	LONG

Cluster 8 – Emerging Markets Equity (19/25)**Members:** 25 **Var explained:** 0.875 **Signal:** LONG since 2026-04-10 (21 days, 9.9%)**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7724
Asia Equity	2	0.0812
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0357
Global Dividend Equity	1	0.0348
South Korea Equity	1	0.0341

Replication basket – IEMG+DEM+FEMS (cor=1.00) basket lead at lag 1: -0.023

ticker	weight_ols	name	category	etf_signal
IEMG	0.754	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG
DEM	0.168	WisdomTree Emerging Markets High Divi...	Emerging Markets Equi...	LONG
FEMS	0.067	First Trust Emerging Markets Small Ca...	Emerging Markets Equi...	FLAT

Members

	ticker	weight	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
16	GEM	0.0423	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
15	EMGF	0.0419	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
18	XSOE	0.0419	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
11	EEMA	0.0413	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
12	PXH	0.0410	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
19	EEMV	0.0404	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG

10	AIA	0.0401	iShares Asia 50 ETF	Asia Equity	LONG
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0390	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0385	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
13	DVYE	0.0368	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0357	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
24	SDIV	0.0348	Global X SuperDividend ETF	Global Dividend Equity	LONG
25	FEMS	0.0347	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	LONG

Cluster 9 – High Yield Bond (10) + Preferred Stock (7)

Members: 22 **Var explained:** 0.772 **Signal:** LONG since 2026-04-13 (20 days, 0.7%)

Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.4918
Preferred Stock	7	0.2893
Emerging Markets Fixed Income	4	0.1695
Unknown	1	0.0493

Replication basket – JNK+PFF+PCY (cor=0.99) basket lead at lag 1: 0.033

ticker	weight_ols	name	category	etf_signal
JNK	0.487	State Street SPDR Bloomberg High Yiel...	High Yield Bond	LONG
PFF	0.257	iShares Preferred and Income Securiti...	Preferred Stock	LONG
PCY	0.115	Invesco Emerging Markets Sovereign De...	Emerging Markets Fixe...	FLAT

Members

	ticker	weight	name	category	etf_signal
14	JNK	0.0502	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
1	HYG	0.0501	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
20	HYLB	0.0500	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
19	BBHY	0.0499	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG

17	SHYG	0.0497	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG
15	SJNK	0.0496	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
8	PHB	0.0493			LONG
16	ANGL	0.0492	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	LONG
18	FALN	0.0492	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0488	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
10	PFF	0.0453	iShares Preferred and Income Securities...	Preferred Stock	LONG
3	HYLS	0.0450	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0444	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
22	VWOB	0.0443	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
12	PFXF	0.0438	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
21	PCY	0.0435	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
7	PGX	0.0421	Invesco Preferred ETF	Preferred Stock	LONG
11	PSK	0.0416	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0410	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0407	Invesco Financial Preferred ETF	Preferred Stock	FLAT
13	HYEM	0.0373	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
9	VRP	0.0348	Invesco Variable Rate Preferred ETF	Preferred Stock	LONG

Cluster 10 – U.S. Dividend Equity (11/20)

Members: 20 **Var explained:** 0.832 **Signal:** LONG since 2026-04-21 (14 days, 0.6%)

Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	11	0.5763
MLP / Energy Infrastructure	4	0.1667
Value Equity	2	0.1037
Dividend Equity	1	0.0524
U.S. Defensive Equity	1	0.0515
U.S. Low Volatility Equity	1	0.0495

Replication basket – DVY+AML+FDL (cor=0.99) basket lead at lag 1: 0.029

ticker	weight_ols	name	category	etf_signal
DVY	0.571	iShares Select Dividend ETF	U.S. Dividend Equity	LONG
AML	0.129	Alerian MLP ETF	MLP / Energy Infrastr...	LONG

FDL 0.263 First Trust Morningstar ETF U.S. Dividend Equity FLAT

Members

	ticker	weight	name	category	etf_signal
15	DVY	0.0538	iShares Select Dividend ETF	U.S. Dividend Equity	LONG
20	SPYD	0.0535	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	LONG
14	SDOG	0.0534	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	LONG
6	DHS	0.0533	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	LONG
19	ONEY	0.0528	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	LONG
11	FTA	0.0526	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	FLAT
5	FDL	0.0524	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
18	SPHD	0.0524	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
10	SCHD	0.0521	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
13	DIV	0.0517	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
16	RDIV	0.0516	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	LONG
7	CDC	0.0515	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	LONG
17	RPV	0.0511	Invesco S&P 500 Pure Value ETF	Value Equity	LONG
12	PEY	0.0510	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	LONG
9	HDV	0.0508	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
8	LVHD	0.0495	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
1	EMLP	0.0472	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
4	MLPX	0.0419	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
2	AMPLP	0.0393	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
3	MLPA	0.0383	Global X MLP ETF	MLP / Energy Infrastruct...	LONG

Cluster 11 – Healthcare (16/17)

Members: 17 **Var explained:** 0.764 **Signal:** LONG since 2026-05-06 (3 days, -2.0%)

Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – FXH+XBI+VHT (cor=0.99) basket lead at lag 1: -0.011

ticker	weight_ols	name	category	etf_signal
FXH	0.363	First Trust Health Care AlphaDEX	Healthcare	FLAT
XBI	0.185	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
VHT	0.472	Vanguard Health Care ETF	Healthcare	FLAT

Members

	ticker	weight	name	category	etf_signal
15	VHT	0.0653	Vanguard Health Care ETF	Healthcare	FLAT
16	FHLC	0.0652	Fidelity MSCI Health Care Index ETF	Healthcare	FLAT
9	FXH	0.0645	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0639	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0637	iShares U.S. Healthcare ETF	Healthcare	FLAT
10	IXJ	0.0622	iShares Global Healthcare ETF	Healthcare	FLAT
1	XLV	0.0620	State Street Health Care Select Sector ...	Healthcare	FLAT
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0595	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0573	iShares U.S. Medical Devices ETF	Healthcare	FLAT
5	PTH	0.0545	Invesco Dorsey Wright Healthcare Moment...	Healthcare	FLAT
17	PPH	0.0542	VanEck Pharmaceutical ETF	Healthcare	FLAT
4	XBI	0.0535	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0511	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0470	iShares U.S. Health Care Providers ETF	Healthcare	FLAT

Cluster 12 – Materials (13/17)**Members:** 17 **Var explained:** 0.756 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Materials	13	0.7907
Consumer Staples	1	0.0589
Energy	1	0.0571
Norway Equity	1	0.0519

Industrial Metals	1	0.0414
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Replication basket – GNR+IYM+PICK (cor=0.99) basket lead at lag 1: 0.041

ticker	weight_ols	name	category	etf_signal
GNR	0.410	State Street SPDR S&P Global Natural ...	Materials	FLAT
IYM	0.369	iShares U.S. Basic Materials ETF	Materials	FLAT
PICK	0.207	iShares MSCI Global Select Metals & M...	Materials	FLAT

Members

	ticker	weight	name	category	etf_signal
11	GNR	0.0653	State Street SPDR S&P Global Natural Re...	Materials	FLAT
6	MXI	0.0648	iShares Global Materials ETF	Materials	FLAT
10	GUNR	0.0643	FlexShares Global Upstream Natural Reso...	Materials	FLAT
7	IYM	0.0641	iShares U.S. Basic Materials ETF	Materials	FLAT
12	PICK	0.0627	iShares MSCI Global Select Metals & Min...	Materials	FLAT
9	VAW	0.0622	Vanguard Materials ETF	Materials	FLAT
17	NANR	0.0614	State Street SPDR S&P North American Na...	Materials	LONG
1	XLB	0.0612	State Street Materials Select Sector SP...	Materials	FLAT
8	RSPM	0.0611	Invesco S&P 500 Equal Weight Materials ...	Materials	FLAT
13	SLX	0.0606	VanEck Steel ETF	Materials	LONG
4	XME	0.0605	State Street SPDR S&P Metals & Mining E...	Materials	LONG
5	MOO	0.0589	VanEck Agribusiness ETF	Consumer Staples	LONG
3	COPX	0.0587	Global X Copper Miners ETF	Materials	LONG
16	IGE	0.0571	iShares North American Natural Resource...	Energy	LONG
14	NORW	0.0519	Global X MSCI Norway ETF	Norway Equity	LONG
2	URA	0.0438	Global X Uranium ETF	Materials	LONG
15	DBB	0.0414	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 13 – China Equity (13/15)

Members: 15 **Var explained:** 0.804 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0642
Hong Kong Equity	1	0.0591

Replication basket – CXSE+CNYY+MCHI (cor=0.99) basket lead at lag 1: -0.040

ticker	weight_ols	name	category	etf_signal
CXSE	0.454	WisdomTree China ex-State-Owned Enter...	China Equity	FLAT
CNYY	0.207	iShares MSCI China A ETF	China Equity	LONG
MCHI	0.320	iShares MSCI China ETF	China Equity	FLAT

Members

	ticker	weight	name	category	etf_signal
11	CXSE	0.0736	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0729	iShares MSCI China ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
2	FXI	0.0711	iShares China Large-Cap ETF	China Equity	FLAT
7	CHIQ	0.0706	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0679	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYY	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0642	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
14	CNXT	0.0595	VanEck ChiNext Innovators ETF	China Equity	LONG
10	KBA	0.0594	KraneShares Boseri MSCI China A 50 Conn...	China Equity	LONG
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	LONG
15	KURE	0.0587	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 14 – Real Estate (14/14)

Members: 14 **Var explained:** 0.962 **Signal:** LONG since 2026-04-10 (21 days, 4.2%)

Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – SCHH+RWR+INDS (cor=1.00) basket lead at lag 1: -0.001

ticker	weight_ols	name	category	etf_signal
SCHH	0.541	Schwab U.S. REIT ETF	Real Estate	LONG
RWR	0.336	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
INDS	0.100	Pacer Industrial Real Estate ETF	Real Estate	LONG

Members

	ticker	weight	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	LONG
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	LONG
10	USRT	0.0723	iShares Core U.S. REIT ETF	Real Estate	LONG
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	LONG
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	LONG
7	REET	0.0715	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0689	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0683	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 15 – Mixed: DBP, GLTR, GDXJ

Members: 13 **Var explained:** 0.831 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
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Materials	5	0.3889
Gold	3	0.2341
Precious Metals	2	0.1624
Silver	2	0.1525
Platinum	1	0.0622

Replication basket – GDJX+GLTR+SIL (cor=1.00) basket lead at lag 1: 0.002

ticker	weight_ols	name	category	etf_signal
GDJX	0.218	VanEck Junior Gold Miners ETF	Materials	FLAT
GLTR	0.601	abrdn Physical Precious Metals Basket...	Precious Metals	FLAT
SIL	0.168	Global X Silver Miners ETF	Materials	LONG

Members

	ticker	weight	name	category	etf_signal
12	DBP	0.0814	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0810	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
2	GDJX	0.0786	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDJ	0.0785	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0781	SPDR Gold Shares	Gold	FLAT
10	IAU	0.0781	iShares Gold Trust Shares	Gold	FLAT
11	SGOL	0.0779	abrdn Physical Gold Shares ETF	Gold	FLAT
13	GOAU	0.0777	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0776	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0765	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0763	iShares Silver Trust	Silver	LONG
9	SIVR	0.0761	abrdn Physical Silver Shares ETF	Silver	LONG
8	PPLT	0.0622	abrdn Physical Platinum Shares ETF	Platinum	LONG

Cluster 16 – Municipal Bond (13/13)

Members: 13 **Var explained:** 0.819 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – VTEB+HYD+CMF (cor=0.99) basket lead at lag 1: 0.188

ticker	weight_ols	name	category	etf_signal
VTEB	0.458	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
HYD	0.172	VanEck High Yield Muni ETF	Municipal Bond	LONG
CMF	0.267	iShares California Muni Bond ETF	Municipal Bond	LONG

Members

	ticker	weight	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0801	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0796	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0694	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0668	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Consumer Staples (6/11)

Members: 11 **Var explained:** 0.748 **Signal:** LONG since 2026-05-01 (6 days, -2.3%)

Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5726
Utilities	4	0.3528

U.S. Small Cap Consumer Staples 1 0.0746

Replication basket – VDC+FUTY+PSCC (cor=1.00) basket lead at lag 1: -0.008

ticker	weight_ols	name	category	etf_signal
VDC	0.527	Vanguard Consumer Staples ETF	Consumer Staples	LONG
FUTY	0.361	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
PSCC	0.085	Invesco S&P SmallCap Consumer Staples...	U.S. Small Cap Consum...	LONG

Members

	ticker	weight	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0973	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0947	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0944	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0892	iShares U.S. Utilities ETF	Utilities	FLAT
8	VPU	0.0880	Vanguard Utilities ETF	Utilities	FLAT
9	FUTY	0.0880	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
2	XLU	0.0875	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0746	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 18 – Mixed: ILF, EWZ, ECH

Members: 11 **Var explained:** 0.325 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Latin America Equity	1	0.1565
Brazil Equity	1	0.1432
Chile Equity	1	0.1273
Malaysia Equity	1	0.1242
Indonesia Equity	1	0.1137

Philippines Equity	1	0.1051
Vietnam Equity	1	0.0792
Agricultural Commodities	1	0.0598
Turkey Equity	1	0.0575
Unknown	1	0.0189
Natural Gas	1	0.0147

Replication basket – ILF+EIDO+ECH (cor=0.96) basket lead at lag 1: 0.018

ticker	weight_ols	name	category	etf_signal
ILF	0.40	iShares Latin America 40 ETF	Latin America Equity	FLAT
EIDO	0.20	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
ECH	0.13	iShares MSCI Chile ETF	Chile Equity	LONG

Members

	ticker	weight	name	category	etf_signal
4	ILF	0.1565	iShares Latin America 40 ETF	Latin America Equity	FLAT
1	EWZ	0.1432	iShares MSCI Brazil ETF	Brazil Equity	FLAT
3	ECH	0.1273	iShares MSCI Chile ETF	Chile Equity	LONG
8	EWM	0.1242	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
6	EIDO	0.1137	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1051	iShares MSCI Philippines ETF	Philippines Equity	FLAT
11	VNM	0.0792	VanEck Vietnam ETF	Vietnam Equity	FLAT
2	DBA	0.0598	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
10	TUR	0.0575	iShares MSCI Turkey ETF	Turkey Equity	LONG
9	RSX	0.0189			LONG
5	UNG	0.0147	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 19 – Inflation-Protected Fixed Income (10/10)

Members: 10 **Var explained:** 0.908 **Signal:** LONG since 2026-04-09 (22 days, 0.8%)

Category breakdown

category	n_members	abs_weight
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Inflation-Protected Fixed Income	10	1
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Replication basket – TIP+STPZ+SPIP (cor=1.00) basket lead at lag 1: -0.019

ticker	weight_ols	name	category	etf_signal
TIP	0.627	iShares TIPS Bond ETF	Inflation-Protected F...	LONG
STPZ	0.194	PIMCO 1-5 Year U.S. TIPS Index Exchan...	Inflation-Protected F...	LONG
SPIP	0.136	State Street SPDR Portfolio TIPS ETF	Inflation-Protected F...	LONG

Members

	ticker	weight	name	category	etf_signal
4	TDTF	0.1033	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1032	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	LONG
3	TIP	0.1028	iShares TIPS Bond ETF	Inflation-Protected Fixe...	LONG
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	LONG
5	TDTT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	LONG
9	STIP	0.0999	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0998	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0997	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	LONG
2	STPZ	0.0995	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG
1	LTPZ	0.0886	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 20 – Japan Equity (7/9)

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7796
Asia-Pacific Equity	2	0.2204

Replication basket – EWJ+DXJ+VPL (cor=1.00) basket lead at lag 1: -0.032

ticker	weight_ols	name	category	etf_signal
EWJ	0.410	iShares MSCI Japan Index Fund	Japan Equity	LONG
DXJ	0.405	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
VPL	0.171	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG

Members

	ticker	weight	name	category	etf_signal
1	EWJ	0.1141	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
9	HEWJ	0.1115	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1114	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1106	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1091	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1082	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 21 – Broad Commodities (6/9)

Members: 9 **Var explained:** 0.913 **Signal:** LONG since 2026-01-08 (84 days, 47.8%)

Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6698
Oil	2	0.2183
Energy Commodities	1	0.1119

Replication basket – GSG+DBO+PDBC (cor=1.00) basket lead at lag 1: 0.057

ticker	weight_ols	name	category	etf_signal
GSG	0.478	iShares GSCI Commodity-Indexed Trust ...	Broad Commodities	FLAT
DBO	0.179	Invesco DB Oil Fund	Oil	FLAT

PDBC	0.353	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG
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Members

	ticker	weight	name	category	etf_signal
5	GSG	0.1151	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1149	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
7	PDBC	0.1149	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
8	COMT	0.1148	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
9	DBE	0.1119	Invesco DB Energy Fund	Energy Commodities	FLAT
1	USO	0.1098	United States Oil Fund	Oil	FLAT
4	DBO	0.1085	Invesco DB Oil Fund	Oil	FLAT
6	DJP	0.1051	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	FTGC	0.1050	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT

Cluster 22 – Mixed: SRLN, BKLN, FTSL**Members:** 9 **Var explained:** 0.560 **Signal:** LONG since 2026-04-07 (24 days, 1.2%)**Category breakdown**

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3632
Interest Rate Hedged Fixed Income	3	0.3245
Floating Rate Fixed Income	3	0.3123

Replication basket – SRLN+LQDH+FLOT (cor=0.97) basket lead at lag 1: -0.033

ticker	weight_ols	name	category	etf_signal
SRLN	0.346	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
LQDH	0.319	iShares Interest Rate Hedged Corporat...	Interest Rate Hedged ...	LONG
FLOT	0.322	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT

Members

	ticker	weight	name	category	etf_signal
2	SRLN	0.1249	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG
3	BKLN	0.1213	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1170	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	LONG
7	FLOT	0.1081	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	LONG
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	LONG
8	FLRN	0.1035	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1008	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 23 – Cash / Ultra-Short Fixed Income (9/9)

Members: 9 **Var explained:** 0.457 **Signal:** FLAT

Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – GSY+NEAR+VNLA (cor=0.95) basket lead at lag 1: 0.026

ticker	weight_ols	name	category	etf_signal
GSY	0.376	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	LONG
NEAR	0.124	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fi...	FLAT
VNLA	0.139	Janus Henderson Short Duration Income...	Cash / Ultra-Short Fi...	LONG

Members

	ticker	weight	name	category	etf_signal
5	GSY	0.1359	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1286	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
7	SHV	0.1261	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1244	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1090	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT

2	MINT	0.1062	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.1032	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.0879	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0787	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 24 – Currency (4/6)**Members:** 6 **Var explained:** 0.742 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Currency	4	0.641
International Fixed Income	2	0.359

Replication basket – IGOV+FXF+FXB (cor=0.98) basket lead at lag 1: 0.036

ticker	weight_ols	name	category	etf_signal
IGOV	0.440	iShares International Treasury Bond E...	International Fixed I...	LONG
FXF	0.292	Invesco CurrencyShares Swiss Franc Tr...	Currency	LONG
FXB	0.203	Invesco CurrencyShares British Pound ...	Currency	LONG

Members

	ticker	weight	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	LONG
1	BWX	0.1789	State Street SPDR Bloomberg Internation...	International Fixed Inco...	LONG
4	FXE	0.1701	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1647	Invesco CurrencyShares Swiss Franc Trust	Currency	LONG
3	FXB	0.1609	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXY	0.1453	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 25 – India Equity (3/4)**Members:** 4 **Var explained:** 0.964 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
India Equity	3	0.7508
Unknown	1	0.2492

Replication basket – INDА+INDY+EPI+PIN (cor=1.00) basket lead at lag 1: -0.071

ticker	weight_ols	name	category	etf_signal
INDA	0.250	iShares MSCI India ETF	India Equity	FLAT
INDY	0.246	iShares India 50 ETF	India Equity	FLAT
EPI	0.256	WisdomTree India Earnings Fund	India Equity	FLAT
PIN	0.249			FLAT

Members

	ticker	weight	name	category	etf_signal
1	INDA	0.2527	iShares MSCI India ETF	India Equity	FLAT
3	EPI	0.2493	WisdomTree India Earnings Fund	India Equity	FLAT
4	PIN	0.2492			FLAT
2	INDY	0.2489	iShares India 50 ETF	India Equity	FLAT

Cluster 26 – Currency (1/2)**Members:** 2 **Var explained:** 0.624 **Signal:** FLAT**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.017

ticker	weight_ols	name	category	etf_signal
SQQQ	0.507	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.496	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Members

ticker	weight	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
RSX			cluster_18	11	LONG	0.371	0.052	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_18	11	FLAT	0.492	0.081	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.692	0.040	0.00007	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.635	0.221	0.00034	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_24	6	FLAT	0.732	0.094	0.00022	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_18	11	FLAT	0.591	0.174	0.00038	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.787	0.130	0.00030	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.825	0.021	0.00000	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.825	0.089	0.00022	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.720	0.182	0.00041	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.656	0.188	0.00047	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_23	9	FLAT	0.793	0.162	0.00038	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_18	11	LONG	0.487	0.219	0.00056	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_23	9	LONG	0.793	0.179	0.00043	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	LONG	0.844	0.124	0.00033	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_24	6	LONG	0.818	0.203	0.00047	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	LONG	0.876	0.160	0.00030	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.836	0.231	0.00044	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.877	0.250	0.00044	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_18	11	FLAT	0.452	0.334	0.00082	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_15	13	LONG	0.750	0.269	0.00072	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.877	0.263	0.00051	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	FLAT	0.804	0.271	0.00070	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_12	17	LONG	0.751	0.357	0.00082	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_18	11	FLAT	0.514	0.374	0.00105	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_18	11	FLAT	0.578	0.409	0.00105	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	LONG	0.855	0.254	0.00070	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_22	9	FLAT	0.864	0.332	0.00070	
ECH	iShares MSCI Chile ETF	Chile Equity	cluster_18	11	LONG	0.637	0.429	0.00116	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	LONG	0.905	0.242	0.00062	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20